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MIL-M-28787/336A(SH)
22 January 1991
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31 JANUARY 1983

MILITARY SPECIFICATION

MODULES, DIGITAL, STANDARD ELECTRONIC

INTERFACE, MICROPROCESSOR MODULE, KEY CODE NBQ

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic interface microprocessor module. This module is class I as specified in MIL-STD-1389.

1.2 Classification. This specification covers two classes of modules. Class I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5963
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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbook. The following specification, standards and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a failure rate predicted in accordance with MIL-STD-1389. The predicted failure rate shall not exceed 0.0760 failures per million operating hours for class I or 0.1757 failures per million operating hours for class II when predicted in accordance with MIL-HDBK-217. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be NBQ.

3.6.2 Size. The module size shall be 1A.

3.6.3 Weight. The module shall have a maximum weight of 35 grams.

3.6.4 Contact pins. The module shall have 40 contact pins positioned 1 through 40.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C		Class I 4/ 60 ± 5°C		Class II 4/ -55 ± 5°C		Class II 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1	Isolation, initial	II	S	10	INF	10	INF	10	INF	10	INF	10	INF	MOhms
	end-of-life	II	S	1	INF	1	INF	1	INF	1	INF	1	INF	MOhms
A2-A16	Isolation	II	S	10	INF	10	INF	10	INF	10	INF	10	INF	MOhms
B	Vcc stress	III	0	---------- Module shall pass all remaining tests ----------										
C1-C13	Vic	IV	S	-0.2	-1.2	-0.2	-1.2	-0.2	-1.2	-0.2	-1.2	-0.2	-1.2	V
D1-D8	Ios	V	X	-15	-83	-15	-83	-15	-83	-15	-83	-15	-83	mA
E1	Iil	VI	X	-1	-820	-1	-820	-1	-820	-1	-820	-1	-820	uA
E2	Iil	VI	X	-1	-1100	-1	-1100	-1	-1100	-1	-1100	-1	-1100	uA
E3-E13	Iil	VI	X	-1	-270	-1	-270	-1	-270	-1	-270	-1	-270	uA
F1	Iih	VII	X	0	30	0	30	0	30	0	30	0	30	uA
F2	Iih	VII	X	0	40	0	40	0	40	0	40	0	40	uA
F3-F13	Iih	VII	X	0	10	0	10	0	10	0	10	0	10	uA
G1-G8	Iozl	VIII	X	-20	20	-20	20	-20	20	-20	20	-20	20	uA
H1-H8	Iozh	IX	X	-20	20	-20	20	-20	20	-20	20	-20	20	uA
J1-J8	Vol	X	X	0	450	0	450	0	450	0	450	0	450	mV
K1-K8	Voh	XI	X	3.5	4.5	3.5	4.5	3.5	4.5	3.5	4.5	3.5	4.5	V
L1-L3	Icc	XII	X	15	145	15	145	15	145	15	145	15	145	mA
M1-M21	Functional	XIII	X	---------- See table XIII ----------										
N1-N8	Propagation delay time, tPHL	XIV	S	5	32	5	40	5	40	5	40	6	40	nS
N9-N32	Propagation delay time, tPHL	XIV	S	5	42	5	50	5	50	5	50	6	50	nS
N33-N40	Propagation delay time, tPHL	XIV	S	5	57	5	65	5	65	5	65	6	65	nS
N41	Propagation delay time, tPHL	XIV	S	5	42	5	50	5	50	5	50	6	50	nS
P1-P8	Propagation delay time, tPLH	XV	S	5	32	5	40	5	40	5	40	6	40	nS

- See footnotes at end of table

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TABLE I. Module electrical requirements - Continued

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Units
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
P3-P32	Propagation delay time, tPLH	XV	5	5	42	5	50	5	50	5	50	6	50	nS
P33,P34	Propagation delay time, tPLH	XV	6	5	32	5	40	5	40	5	40	6	40	nS
Q1-Q24	Disable delay time, tPHZ	XVI	5	5	150	5	150	5	160	5	150	5	165	nS
R1-R24	Disable delay time, tPLZ	XVII	5	5	60	5	60	5	70	5	60	5	75	nS
S1-S24	Enable delay time, tPZL	XVIII	6	5	60	5	60	5	70	5	60	5	75	nS
T1-T24	Enable delay time, tPZH	XIX	5	5	60	5	60	5	70	5	60	5	75	ns
U1-U8	Output crosstalk, low level	XX	0	0	950	0	950	0	950	0	950	0	950	mV
V1-V8	Output crosstalk, high level	XXI	0	0	400	0	400	0	400	0	400	0	400	mV
W	PSTA	XXII	0	0	400	0	400	0	400	0	400	0	400	mV
X	PSFC	XXIII	5	3.3	6.1	3.3	6.1	3.3	6.1	3.3	6.1	3.3	6.1	uF
Y1,Y2	Cin	XXIV	0	2	18	2	18	2	18	2	18	2	18	pF
Y3-Y13	Cin	XXIV	0	2	15	2	15	2	15	2	15	2	15	pF
Z1-Z8	Cout	XXV	0	2	18	2	18	2	18	2	18	2	18	pF
AA1-AA8	Output crosstalk duration, low level	XXVI	0	0	400	0	400	0	400	0	400	0	400	mV

- See footnotes on next page .

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/336-1 for class I and M28787/336-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 $\pm$ 0.5 V dc, except the output short circuit current limits apply only for a power supply voltage of 5.5 V dc $\pm$ 1 percent.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

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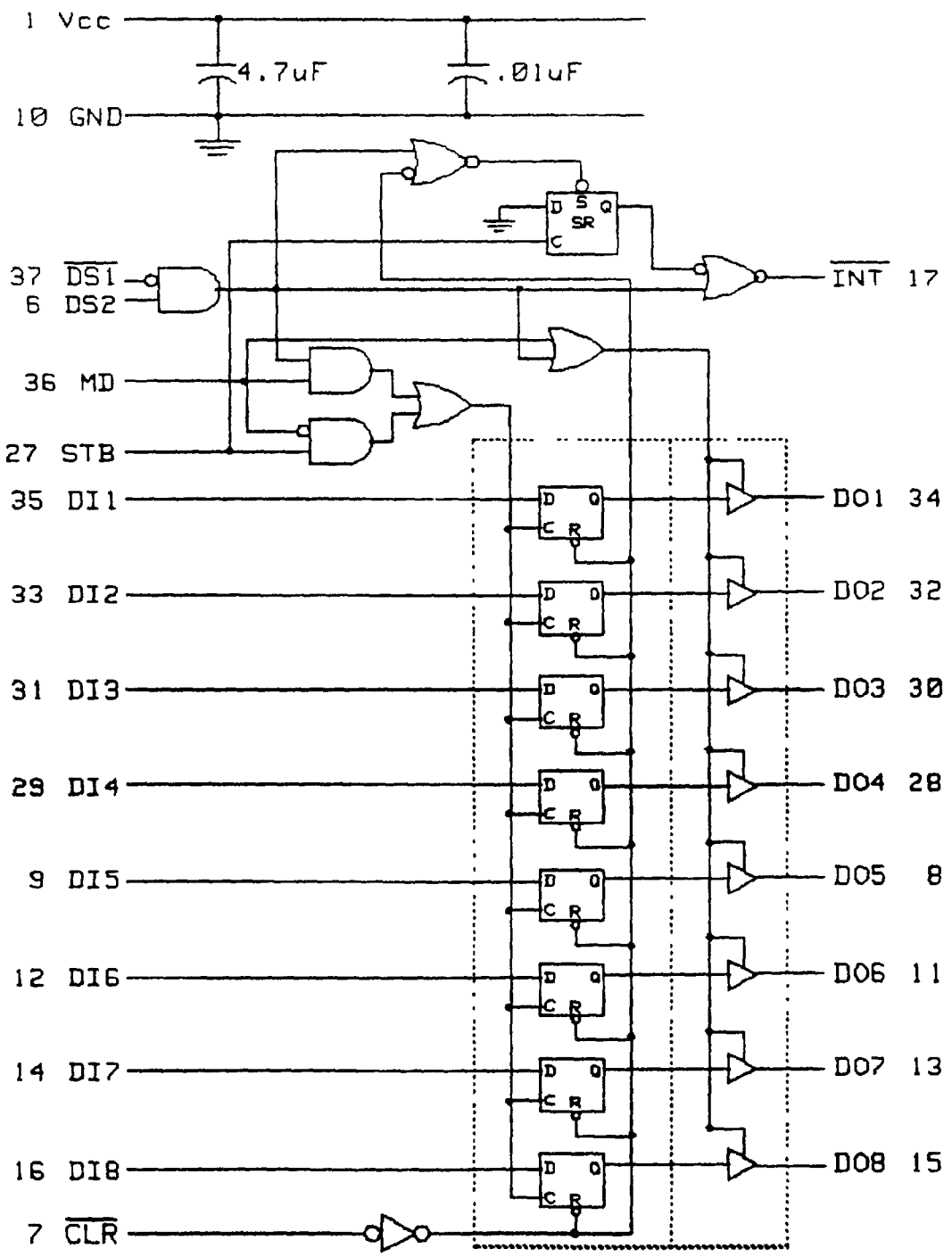


FIGURE 1. NBQ functional block diagram.

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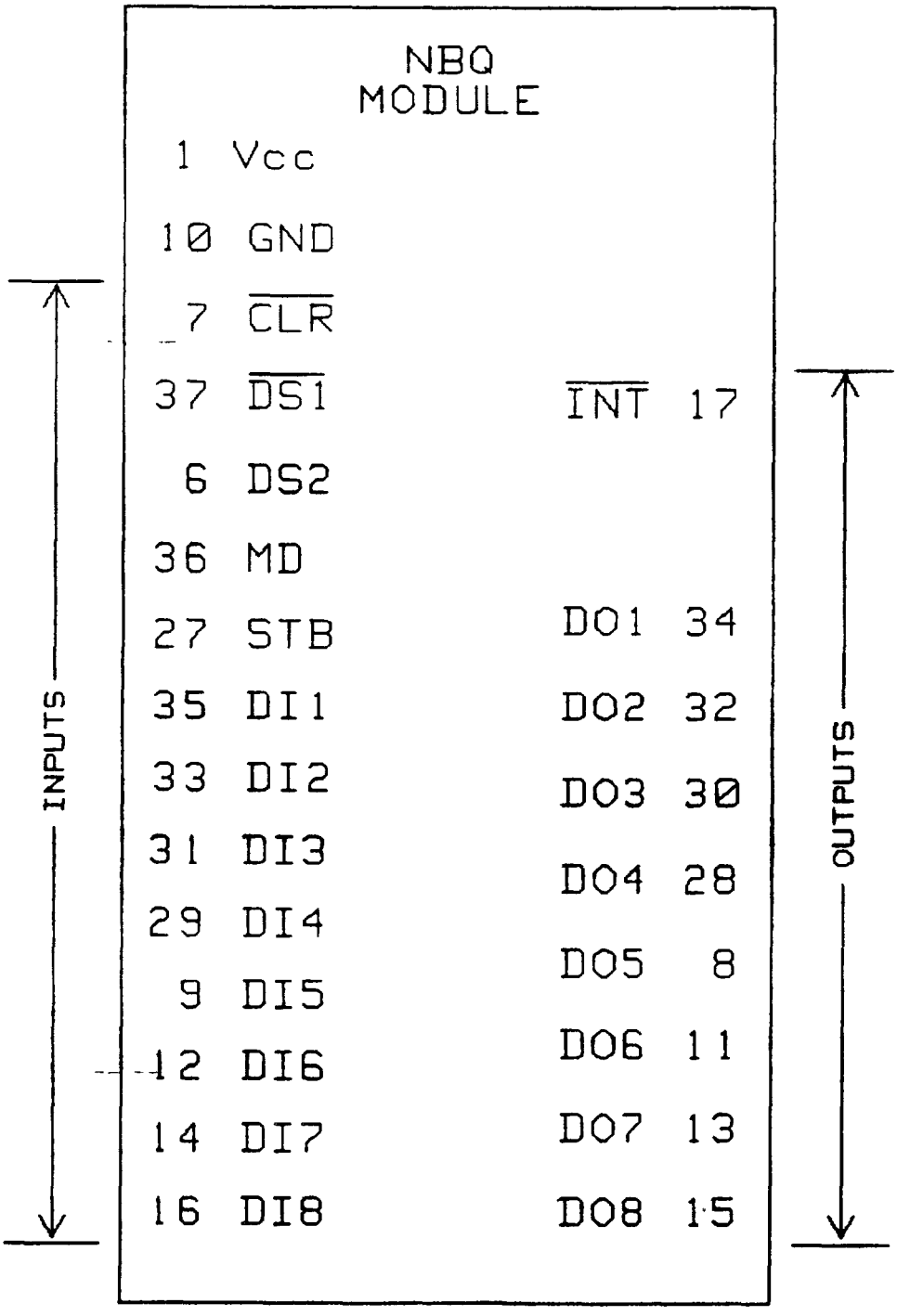


FIGURE 2. NBQ pin assignment.

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4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through AA shall be performed in any order with the following exceptions. Test conditions A, B, C and D are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C and D where applicable, followed by the remaining tests in the sequence.

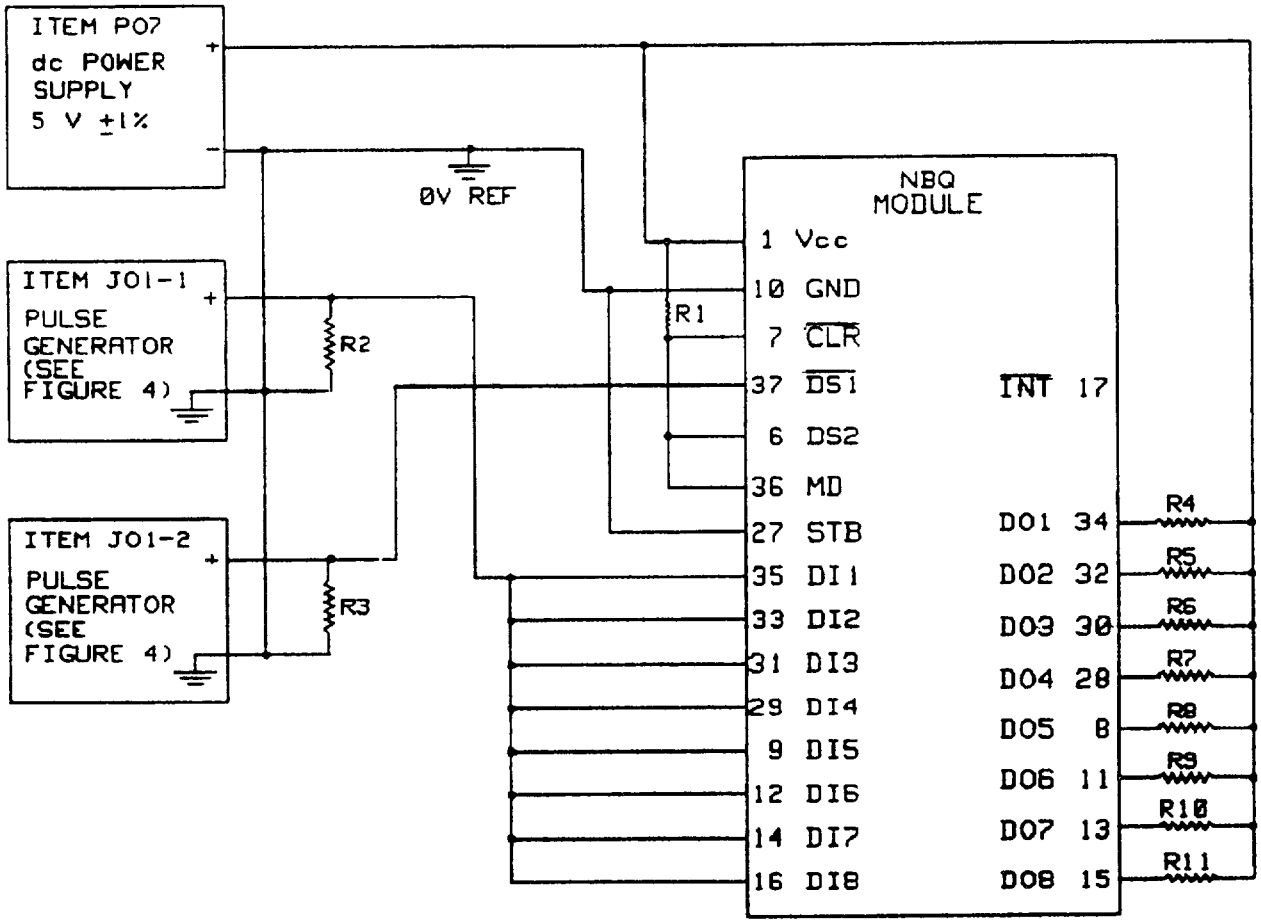
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through AA in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through Y shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated as shown on figures 3 and 4.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 6. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition Y, shall be performed using the initial limits at 25 ± 5°C ambient.

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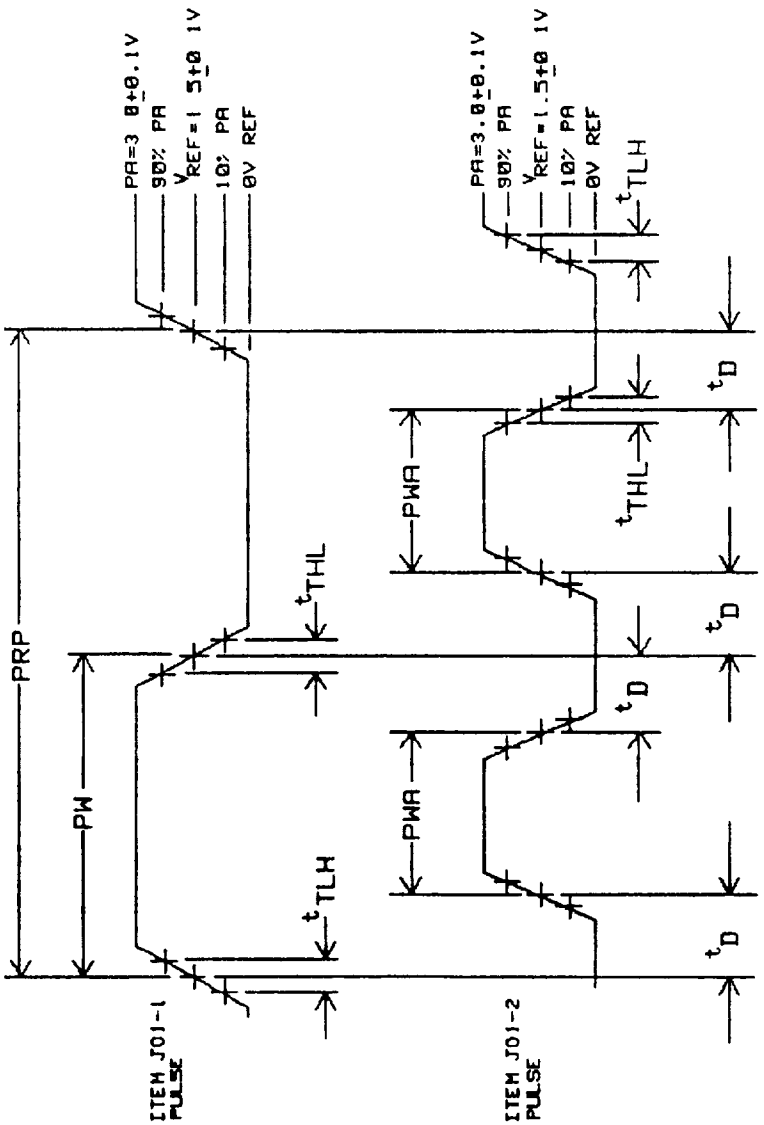


NOTES:

1. R1 = 2.2 KOhms ± 20 percent, 1/4 W.
2. R2 = R3 = 51 Ohms ± 5 percent, 1/2 W.
3. R4 = R11 = 455 Ohms ± 1 percent, 1/8 W.

FIGURE 3. NBQ operating temperature and life test circuit.

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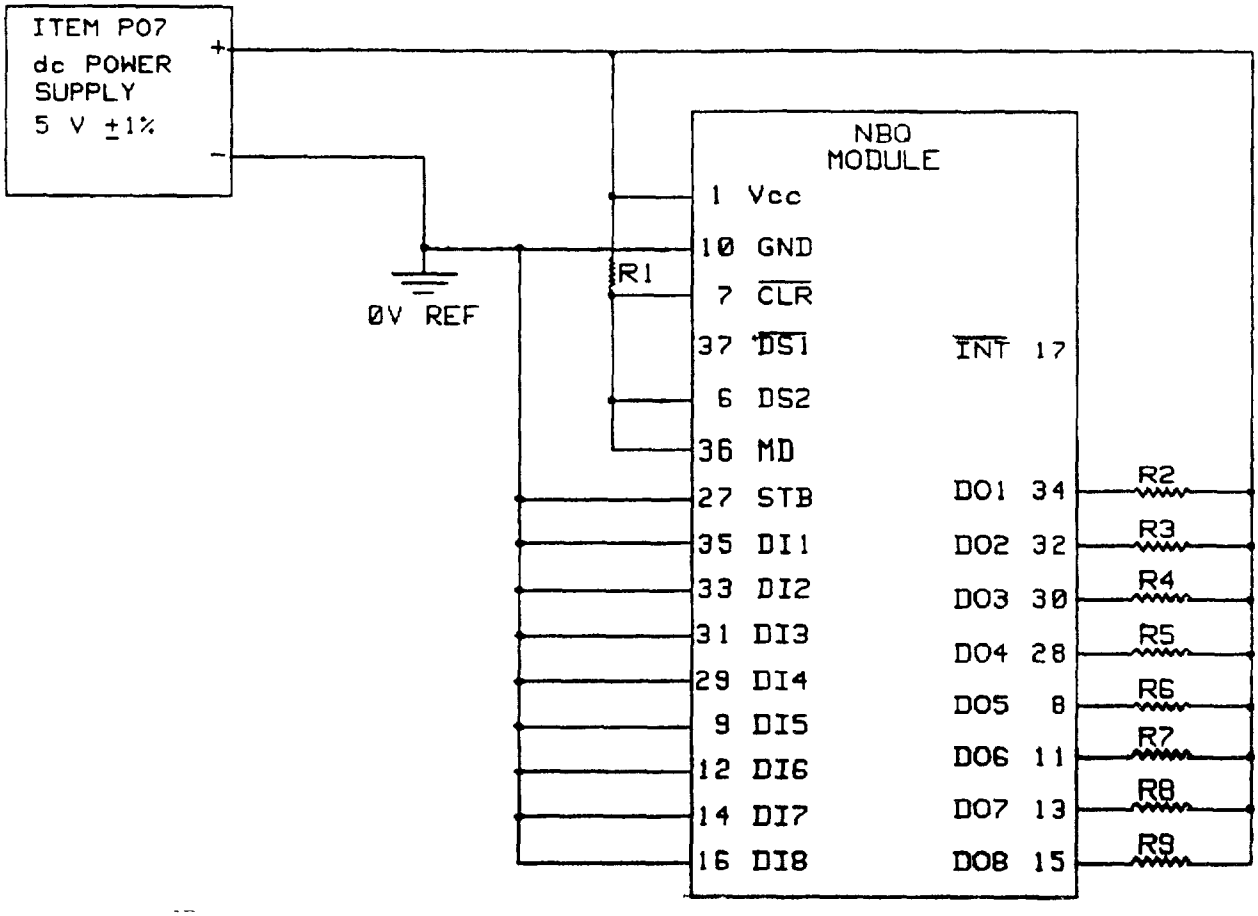
NOTE:

1. Input pulse parameters:

   PW = 500 ± 50 nS.
   PRP = 1000 ± 100 nS.
   tTLH = tTHL = 10 ± 2 nS.
   PWA = 300 ± 30 nS.
   tD = 100 ± 10 nS.

FIGURE 4. Operating temperature and life test waveforms.

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NOTES:

1. R1 = 2.2 KOhms ± 20 percent, 1/4 W.
2. R2 - R9 = 455 Ohms ± 1 percent, 1/8 W.

FIGURE 5. NBQ component temperature test circuit.

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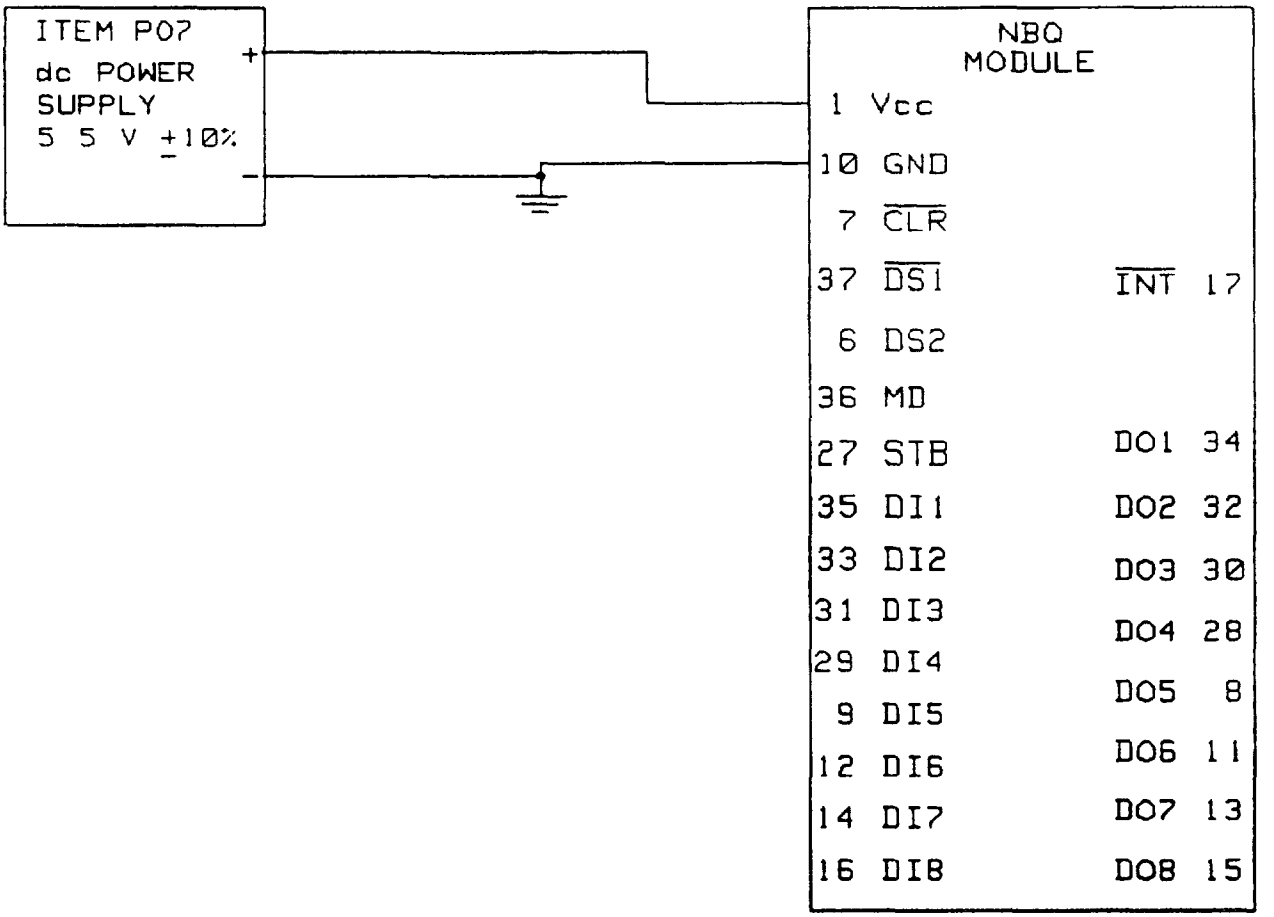


FIGURE 6. NBQ vibration test circuit.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 3 and 4. Following the life test and while still at the maximum class temperature, electrical tests E through Y shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests E through Y shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Cout = output capacitance.
(c) Ibd = input breakdown current.
(d) Icc = power supply current.
(e) Iih = input high level current.
(f) Iil = input low level current.
(g) Iin = input forcing current.
(h) Ioh = output high level current.
(i) Iol = output low level current.
(j) Ios = output short circuit current.
(k) Iozh = output high level leakage current.
(l) Iozl = output low level leakage current.
(m) NA = not applicable.
(n) PSFC = power supply filter capacitance.
(o) PSTA = power supply transient amplitude.
(p) Vcc = Power supply voltage.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 798 mW (max); 433 mW (typical).

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6.3.1 Input parameters.

Signal	Vic max (V)	Vil max (V)	Vih min (V)	Iil max (uA)	Iih max (uA)	Ibd max (mA)	Cin max (pF)
MD	-1.2	0.8	2	-820	30	NA	18
STB	-1.2	0.8	2	-270	10	NA	15
DI1	-1.2	0.8	2	-270	10	NA	15
DI2	-1.2	0.8	2	-270	10	NA	15
DI3	-1.2	0.8	2	-270	10	NA	15
DI4	-1.2	0.8	2	-270	10	NA	15
DI5	-1.2	0.8	2	-270	10	NA	15
DI6	-1.2	0.8	2	-270	10	NA	15
DI7	-1.2	0.8	2	-270	10	NA	15
DI8	-1.2	0.8	2	-270	10	NA	15
$\overline{\text{CLK}}$	-1.2	0.8	2	-270	10	NA	15
$\overline{\text{DS1}}$	-1.2	0.8	2	-1100	40	NA	18
DS2	-1.2	0.8	2	-270	10	NA	15

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (uA)	Iol max (uA)	Iozh max (uA)	Iozl min (uA)	Cout max (pF)
$\overline{INT}$	3.5	0.45	-500	10	NA	NA	NA
DO1	3.5	0.45	-500	10	20	-20	18
DO2	3.5	0.45	-500	10	20	-20	18
DO3	3.5	0.45	-500	10	20	-20	18
DO4	3.5	0.45	-500	10	20	-20	18
DO5	3.5	0.45	-500	10	20	-20	18
DO6	3.5	0.45	-500	10	20	-20	18
DO7	3.5	0.45	-500	10	20	-20	18
DO8	3.5	0.45	-500	10	20	-20	18

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6.4 Module function. The NBQ is a microprocessor interface module. The strobe line (STB) clocks in the data which is activated onto DI1-DI8. Asserting the two data select lines ($\overline{DS1}$ and DS2) enables the eight output transceivers, which activates the data output lines. The NBQ also asserts an interrupt flag when data is clocked into the module. This interrupt flag is output to the interrupt line ($\overline{INT}$) when data is activated onto the output (DO1-DO8) lines. The NBQ also has a clear line ($\overline{CLR}$), which resets the data input latches and the interrupt flag, and disables the output data transceivers. Below is a summary of the NBQ function.

STB	MD	$\overline{DS1}$, DS2	Data Out Equals
0	0	0	3 State
1	0	0	3 State
0	1	0	Data latch
1	1	0	Data latch
0	0	1	Data latch
1	0	1	Data in
0	1	1	Data in
1	1	1	Data in

$\overline{CLR}$	($\overline{DS1}$, DS2)	STB
0	0	0
0	1	0
1	1	↓
1	1	0
1	0	0
1	1	↑

$\overline{INT}$
1
0
0
0
1
0

6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II, figure 7.

(b) Vcc stress test - table III, figure 8.

(c) Vic test - table IV, figure 9.

(d) Ios test - table V, figure 10.

(e) Iil test - table VI, figure 11.

(f) Iih test - table VII, figure 11.

(g) Iozl test - table VIII, figure 12.

(h) Iozh test - table IX, figure 12.

(i) Vol test - table X, figures 13 and 14.

(j) Voh test - table XI, figures 13 and 14.

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(k) Icc test - table XII, figure 15.

(l) Functional test - table XIII, figure 16.

(m) Propagation delay time tests - tables XIV and XV, figures 17 through 22.

(n) Disable/Enable delay time tests - tables XVI, XVII, XVIII and XIX, figures 23 and 24.

(o) Output crosstalk tests - tables XX and XXI, figures 23 and 25.

(p) Output crosstalk duration test - table XXII, figures 23 and 26.

(q) PSTA test - table XXIII, figures 23 and 27.

(r) PSFC test - table XXIV, figure 28.

(s) Cin test - table XXV, figure 28.

(t) Cout test - table XXVI, figure 28.

Preparing activity:
Navy - SH

(Project 5963-N828)

Agent: NW

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TABLE II. Isolation test.

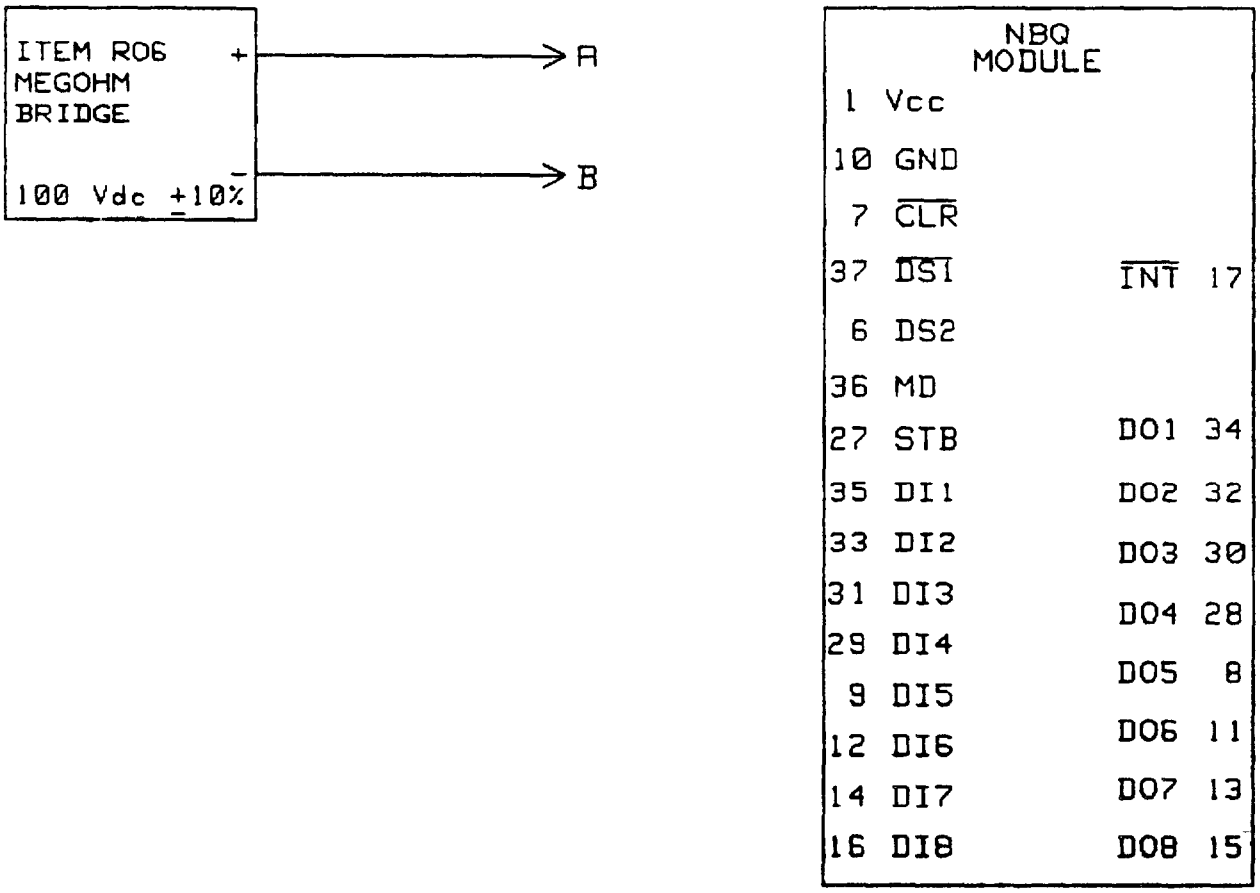
Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
A1	Isolation, frame	All pins	Frame	10	INF	MOhms	7
A2	Isolation, unused to unused	2,3,4,5,18, 19,20,21,22, 23,24,25,26, 38,39	40	10	INF	MOhms	7
A3	Isolation, unused to unused	2,3,4,5,18, 19,20,21,22, 23,24,25,26, 38	39				
A4	Isolation, unused to unused	2,3,4,5,18, 19,20,21,22, 23,24,25,26	38				
A5	Isolation, unused to unused	2,3,4,5,18, 19,20,21,22, 23,24,25	26				
A6	Isolation, unused to unused	2,3,4,5,18, 19,20,21,22, 23,24	25				
A7	Isolation, unused to unused	2,3,4,5,18, 19,20,21,22, 23	24				
A8	Isolation, unused to unused	2,3,4,5,18, 19,20,21,22	23				
A9	Isolation, unused to unused	2,3,4,5,18, 19,20,21	22				
A10	Isolation, unused to unused	2,3,4,5,18, 19,20	21	10	INF	Mohms	7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
A11	Isolation, unused to unused	2,3,4,5,18, 19	20	10	INF	MOhms	7
A12	Isolation, unused to unused	2,3,4,5,18	19				
A13	Isolation, unused to unused	2,3,4,5	18				
A14	Isolation, unused to unused	2,3,4	5				
A15	Isolation, unused to unused	2,3	4				
A16	Isolation, unused to unused	2	3	10	INF	MOhms	7

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NOTES:

1. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

2. When testing for test condition A1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

3. The minimum end-of-life limit is 1 MOhm.

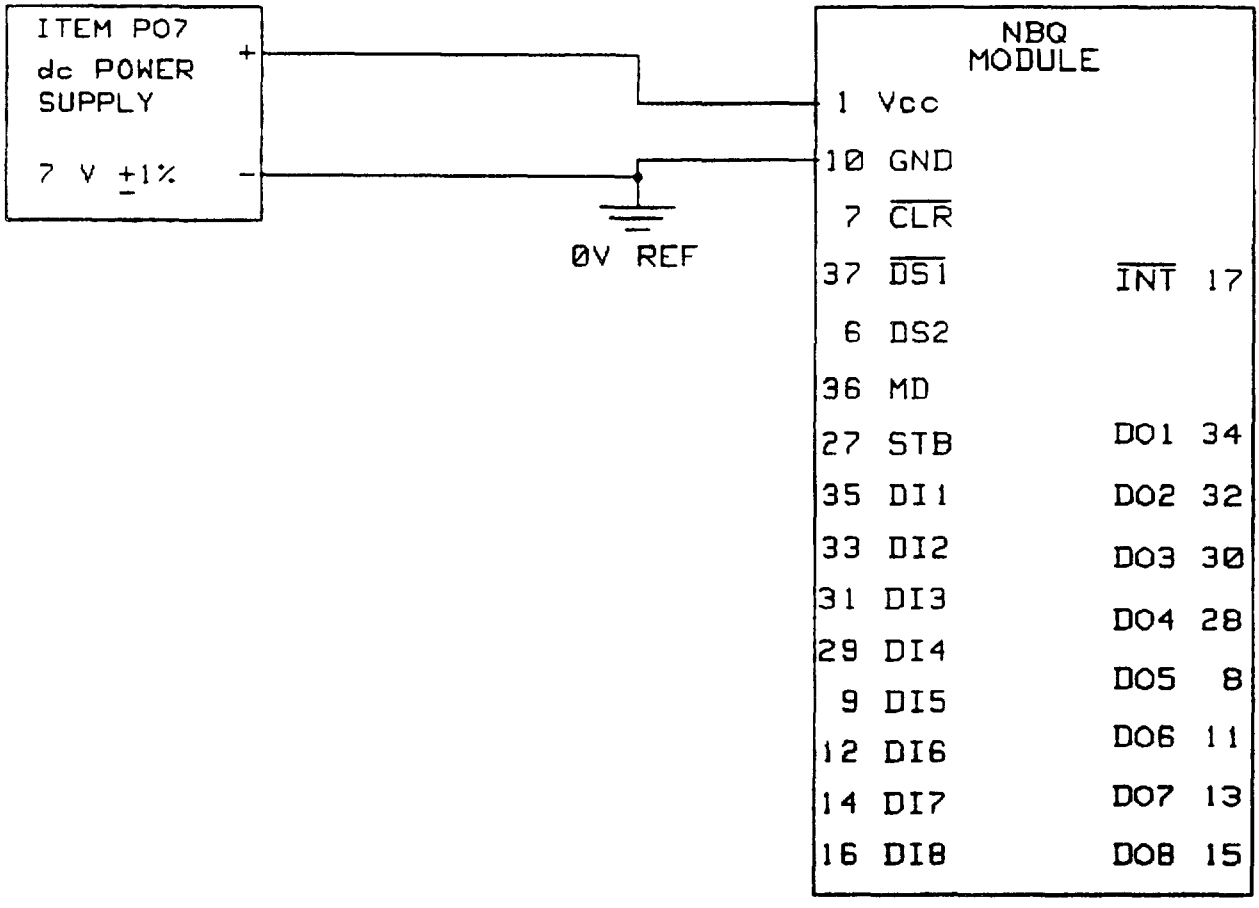
FIGURE 7. Isolation test circuit for table II.

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TABLE III. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect		25 ± 5°C Initial limits	Fig
			Positive lead to	Negative lead to		
B	Vcc	7.0	1	10	Module shall pass all remaining tests.	8

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NOTE:

1. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

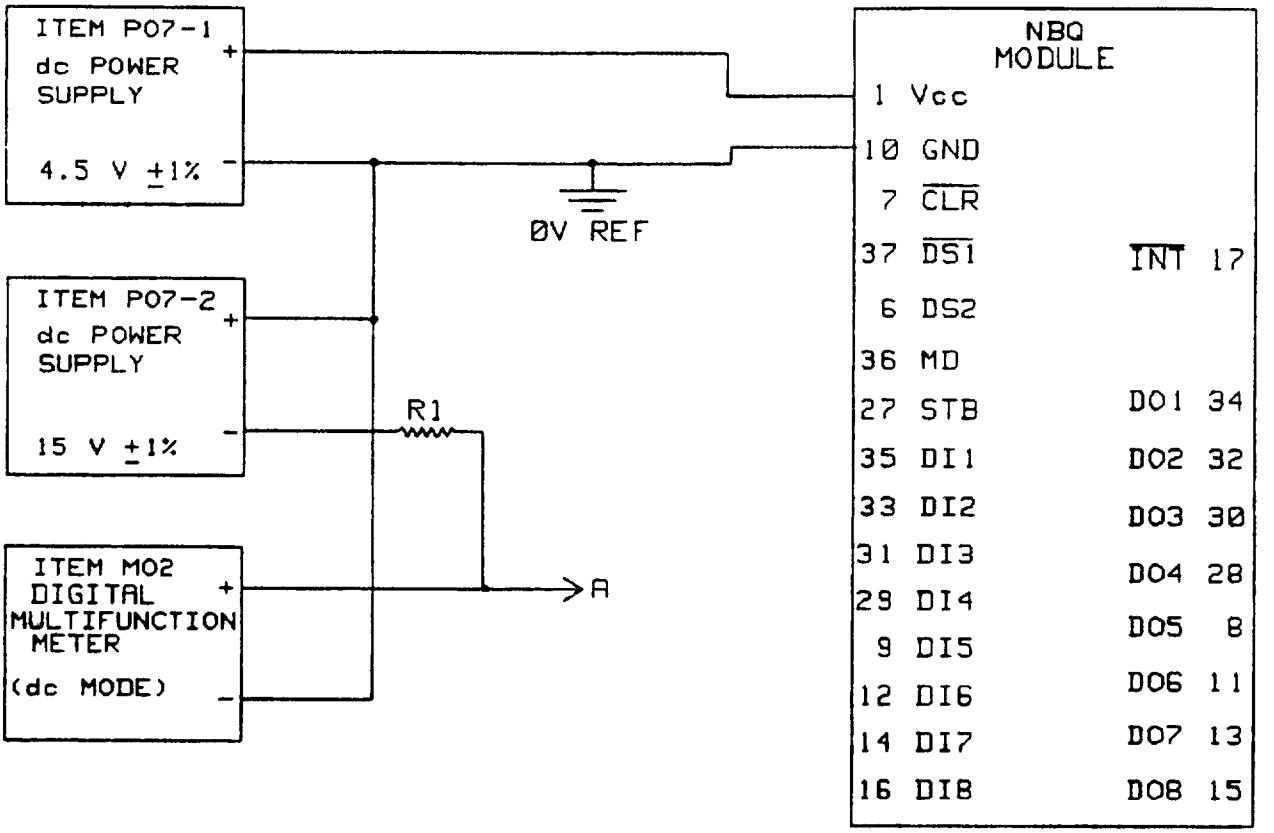
FIGURE 8. Vcc stress test circuit for table III.

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TABLE IV. Input clamp diode voltage test.

Test cond	Signal name	Connect A to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
C1	MD	36	-0.2	-1.2	V	9
C2	$\overline{DS1}$	37				
C3	DS2	6				
C4	$\overline{CLR}$	7				
C5	STB	27				
C6	DI1	35				
C7	DI2	33				
C8	DI3	31				
C9	DI4	29				
C10	DI5	9				
C11	DI6	12				
C12	DI7	14				
C13	DI8	16	-0.2	-1.2	V	9

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:51Z
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NOTE:

1. R1 = 3.01 KOhms ± 1 percent, 1/4 W.

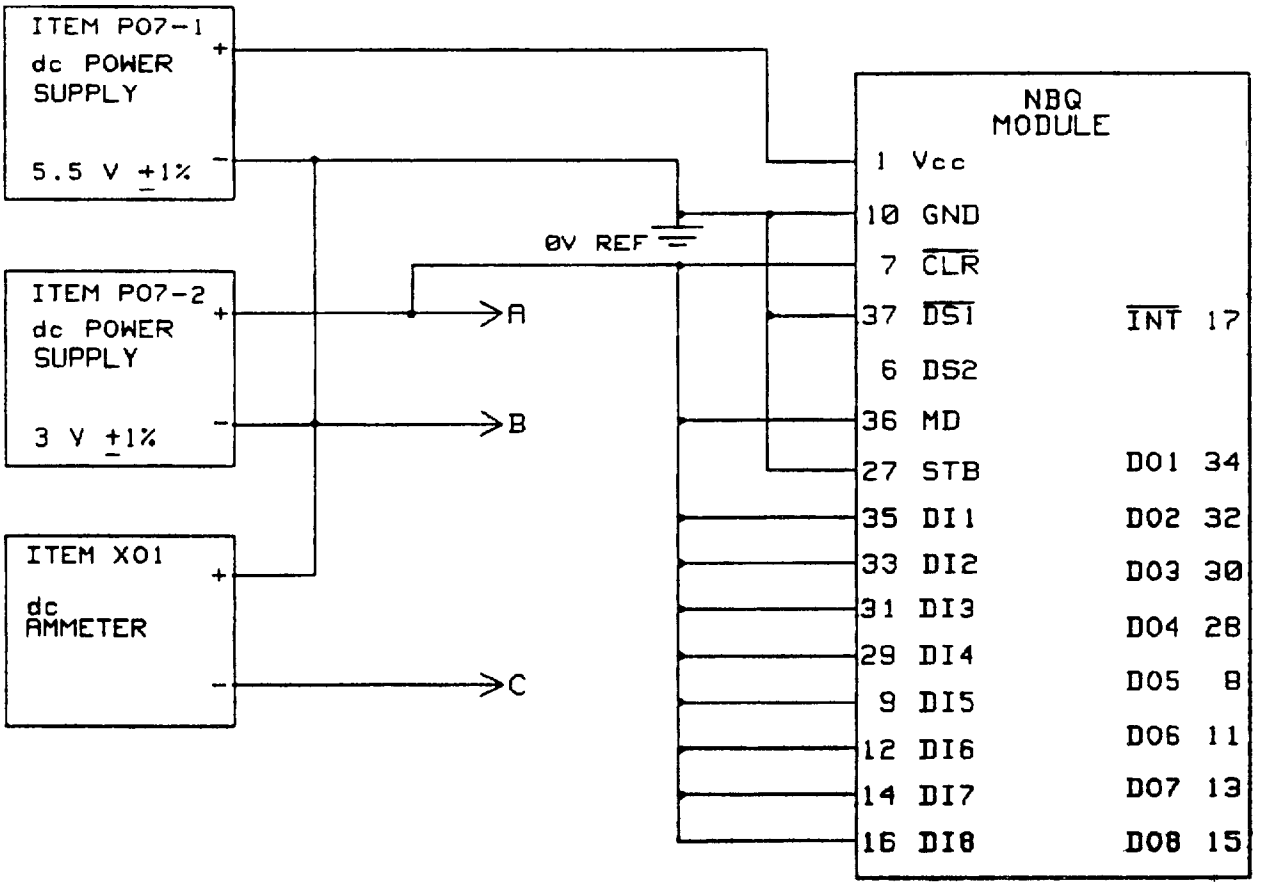
FIGURE 9. Input clamp diode voltage test circuit for table IV.

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TABLE V. Output short circuit current test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
D1	DO1	6	Open	34	-15	-83	mA	10
D2	DO2			32				
D3	DO3			30				
D4	DO4			28				
D5	DO5			8				
D6	DO6			11				
D7	DO7			13				
D8	DO8	6	Open	15				
D9	$\overline{INT}$	Open	6	17	-15	-83	mA	10

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NOTES:

1. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

2. Caution: Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 10. Output short circuit current test circuit for table V.

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TABLE VI. Input current, low level, test.

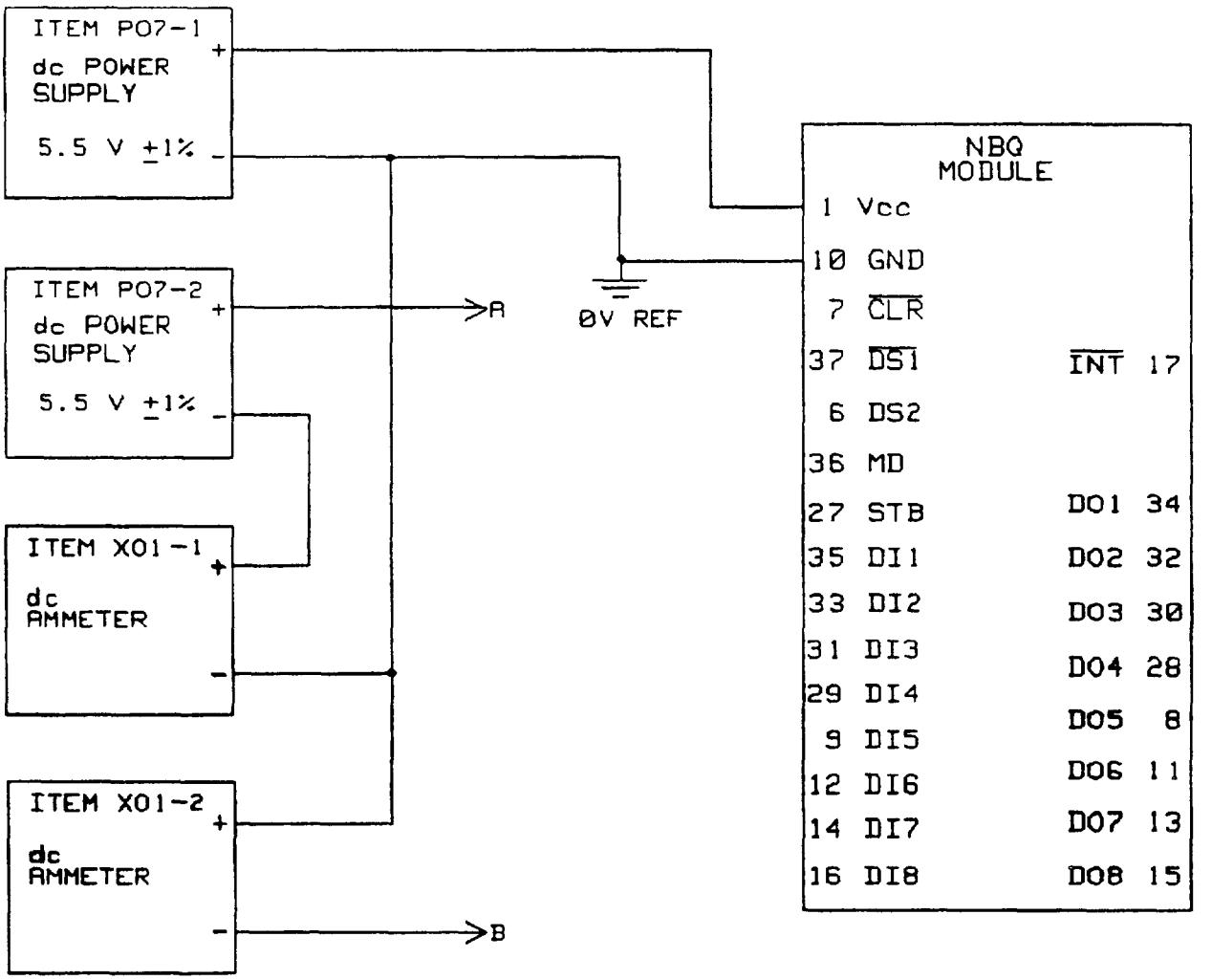
Test cond	Signal name	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
E1	MD	Open	36	-1	-820	uA	11
E2	$\overline{DS1}$		37	-1	-1100	uA	11
E3	DS2		6	-1	-270	uA	11
E4	$\overline{CLR}$		7				
E5	STB		27				
E6	DI1		35				
E7	DI2		33				
E8	DI3		31				
E9	DI4		29				
E10	DI5		9				
E11	DI6		12				
E12	DI7		14				
E13	DI8	Open	16	-1	-270	uA	11

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Connect A to no.	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
F1	MD	36	Open	0	30	uA	11
F2	$\overline{DS1}$	37		0	40	uA	11
F3	DS2	6		0	10	uA	11
F4	$\overline{CLR}$	7					
F5	STB	27					
F6	DI1	35					
F7	DI2	33					
F8	DI3	31					
F9	DI4	29					
F10	DI5	9					
F11	DI6	12					
F12	DI7	14					
F13	DI8	16	Open	0	10	uA	11

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NOTE:

1. Caution: Do not isolate the chassis of ammeter XO1 from earth ground.

FIGURE 11. Input current, low and high level, test circuit for tables VI and VII.

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TABLE VIII. Output leakage current, low level, test.

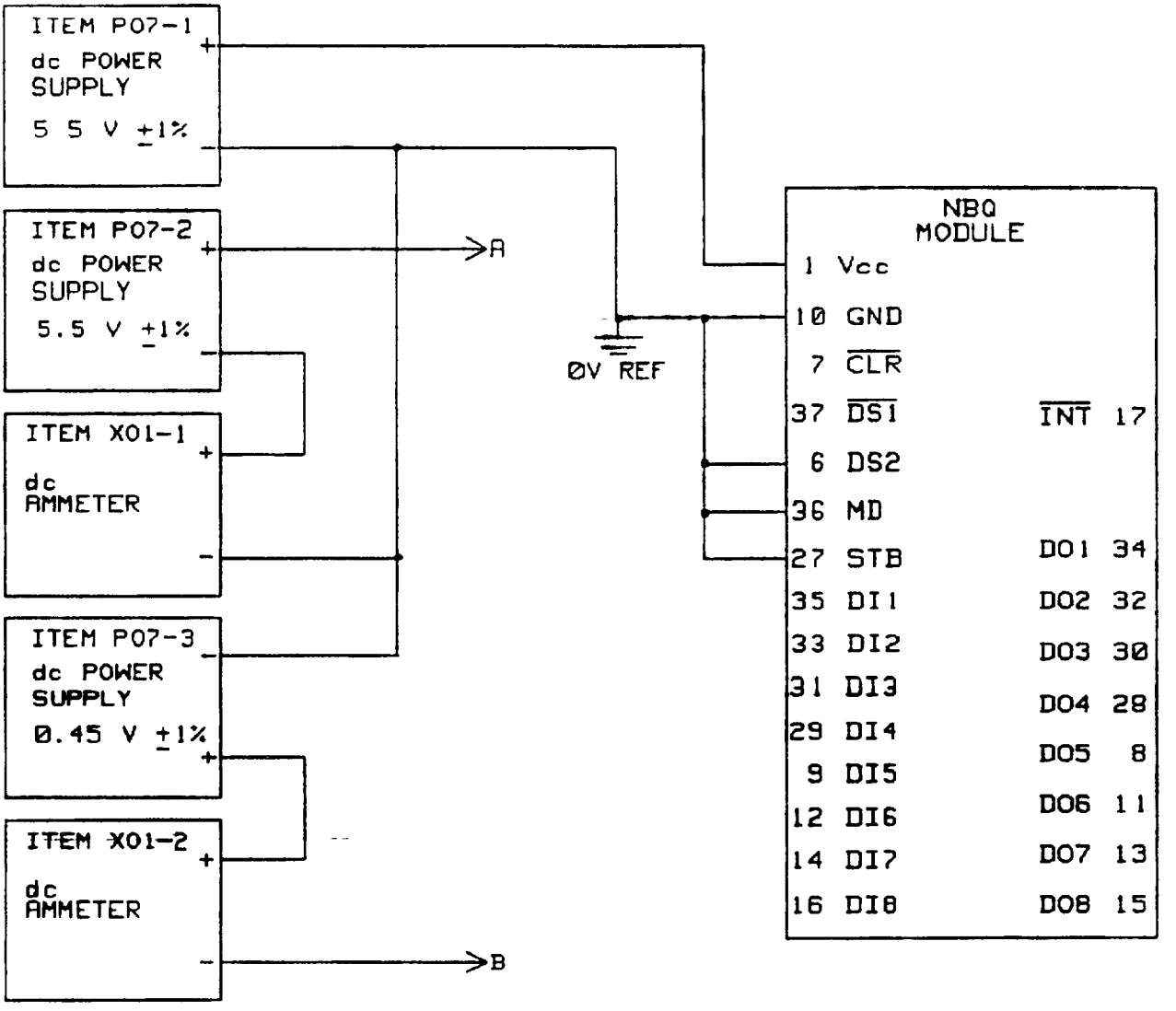
Test cond	Signal name	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
G1	DO1	Open	34	-20	20	uA	12
G2	DO2		32				
G3	DO3		30				
G4	DO4		28				
G5	DO5		8				
G6	DO6		11				
G7	DO7		13				
G8	DO8	Open	15	-20	20	uA	12

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TABLE IX. Output leakage current, high level, test.

Test cond	Signal name	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
H1	DO1	34	Open	-20	20	uA	12
H2	DO2	32					
H3	DO3	30					
H4	DO4	28					
H5	DO5	8					
H6	DO6	11					
H7	DO7	13					
H8	DO8	15	Open	-20	20	uA	12

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NOTE:

1. Caution: Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 12. Output leakage current, low and high level, test circuit for tables VIII and IX.

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TABLE X. Output voltage, low level, test.

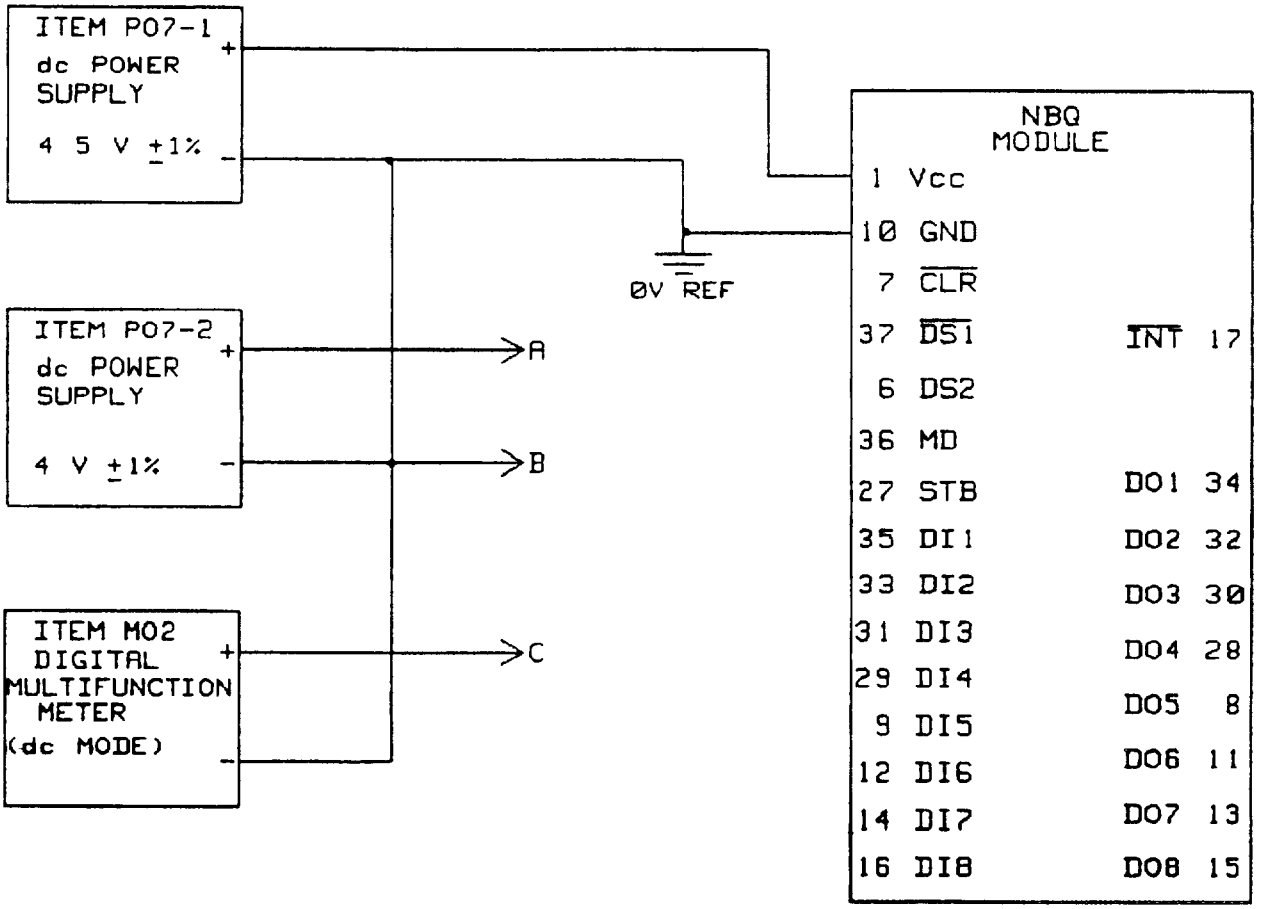
Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
J1	DO1	7,36,6	37,27, 35,33, 31,29, 14,16, 9,12	34	0	450	mV	13
J2	DO2			32				
J3	DO3			30				
J4	DO4			28				
J5	DO5			8				
J6	DO6			11				
J7	DO7			13				
J8	DO8	7,36,6	37,27, 35,33, 31,29, 14,16, 9,12	15				
J9	$\overline{INT}$	6	7,27, 37	17	0	450	mV	13

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TABLE XI. Output voltage, high level, test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
K1	DO1	7,36, 6,35, 33,31, 29,9, 12,14, 16	37,27	34	3.5	4.5	V	13
K2	DO2			32				
K3	DO3			30				
K4	DO4			28				
K5	DO5			8				
K6	DO6			11				
K7	DO7			13				
K8	DO8	7,36, 6,35, 33,31, 29,9, 12,14, 16	37,27	15				
K9	$\overline{INT}$	7,36	37,27, 6	17	3.5	4.5	V	13

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Check the source to verify that this is the current version before use.

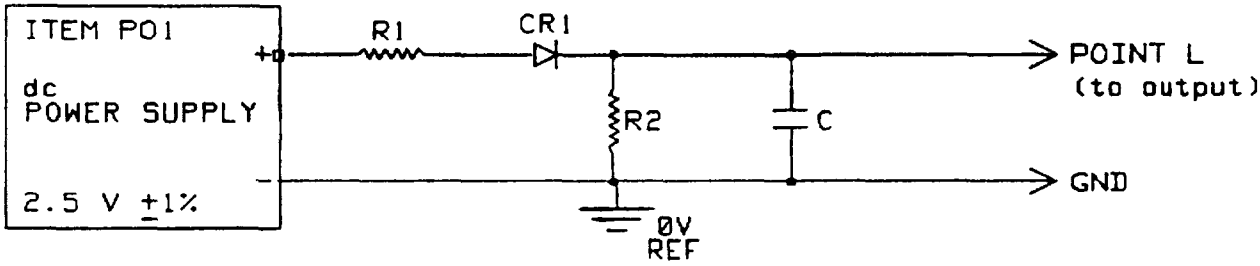


NOTE:

1. Load all outputs as shown on figure 14.

FIGURE 13. Output voltage, low and high level, test circuit for tables X and XI.

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NOTES:

1. CR1 = 1N4150 or equivalent.

2. C = 50 ± 5 pF.

3. Select R1 for a current flow of 10 mA ± 1 percent out of point L with point L held at 0.45 V.

4. R2 = 6.81 KOhms ± 1 percent, 1/8 W.

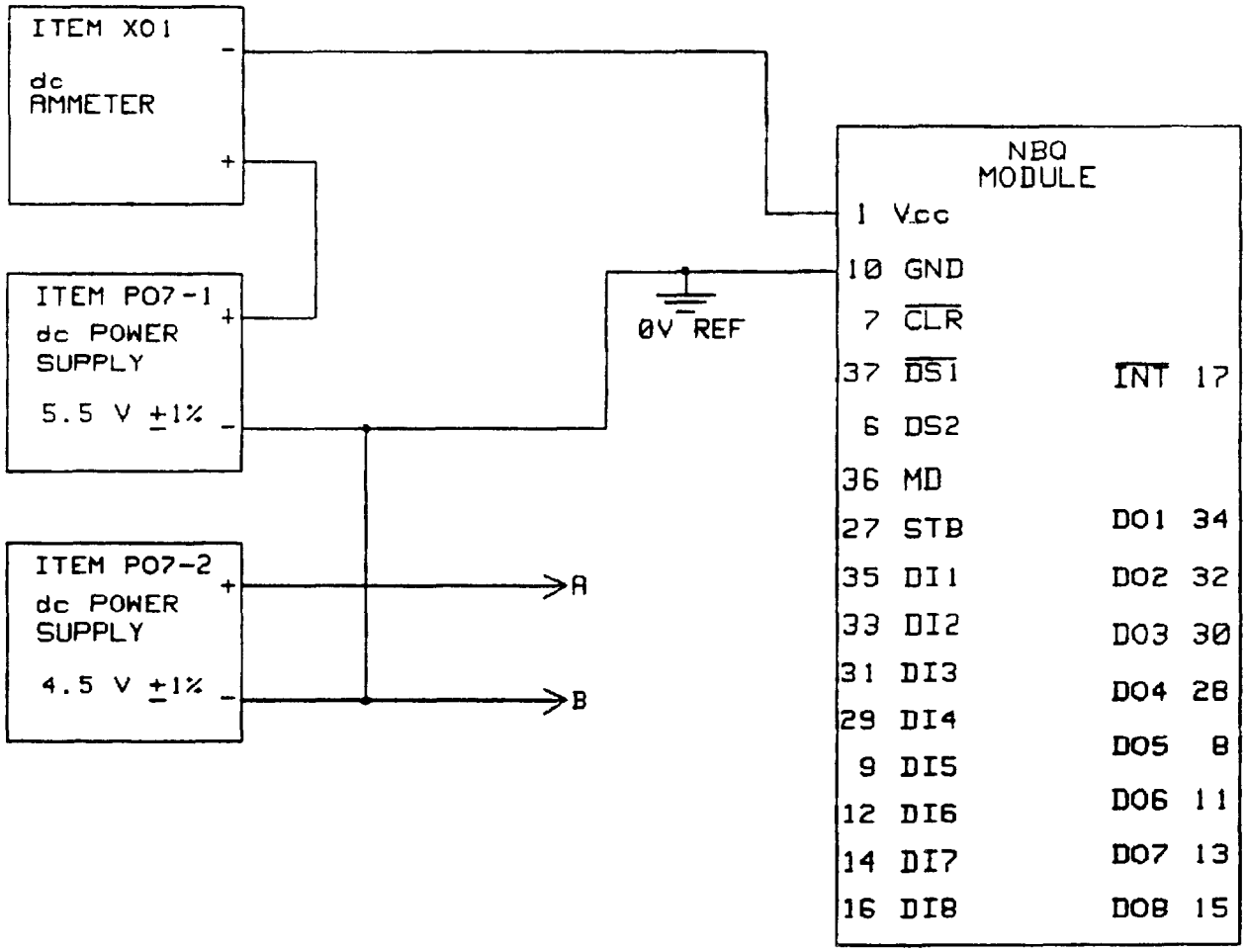
FIGURE 14. Logic load circuit.

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TABLE XII. Power supply current test.

Test cond	Parameter	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
L1	Icc	6	7,37	15	145	mA	15
L2	Icc	6,7,36,35, 33,31,29, 9,12,14,16	37				
L3	Icc	Open	6,36	15	145	mA	15

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NOTE:

1. Caution: Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 15. Power supply current test circuit for table XII.

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TABLE XIII. Functional test.

Test cond	Parameter	Connect										Fig
		A to pin	B to pin	C to pin								
				17	34	32	30	28	0	11	13	15	
M1	Functional	37,36	6,27,7	1	0	0	0	0	0	0	0	0	16
M2	Functional	7,6,27,35	37,36,33,31,29,9,12,14,16	0	1	0	0	0	0	0	0	0	16
M3	Functional	7,6,33,31,29,9,12,14,16	37,36,27,35	0	1	0	0	0	0	0	0	0	16
M4	Functional	7,6,36,33	37,27,35,31,29,9,12,14,16	0	0	1	0	0	0	0	0	0	16
M5	Functional	7,37,6,36,35,31,29,9,12,14,16	27,33	1	0	1	0	0	0	0	0	0	16
M6	Functional	7,6,36,27,31	37,35,33,29,9,12,14,16	0	0	0	1	0	0	0	0	0	16
M7	Functional	7,36,27,35,33,29,9,12,14,16	37,6,31	1	0	0	1	0	0	0	0	0	16
M8	Functional	7,6,27,29	37,36,35,33,31,9,12,14,16	0	0	0	0	1	0	0	0	0	16
M9	Functional	7,6,35,33,31,9,12,14,16	37,36,27,29	0	0	0	0	1	0	0	0	0	16
M10	Functional	7,6,36,9	37,27,35,33,31,29,12,14,16	0	0	0	0	0	1	0	0	0	16
M11	Functional	7,37,6,36,35,33,31,29,12,14,16	27,9	1	0	0	0	0	1	0	0	0	16
M12	Functional	7,6,36,27,12	37,35,33,31,29,9,14,16	0	0	0	0	0	0	1	0	0	16
M13	Functional	7,36,27,35,33,31,29,9,14,16	37,6,12	1	0	0	0	0	0	1	0	0	16
M14	Functional	7,6,27,14	37,36,35,33,31,29,9,12,16	0	0	0	0	0	0	0	1	0	16
M15	Functional	7,6,35,33,31,29,9,12,16	37,36,27,14	0	0	0	0	0	0	0	1	0	16
M16	Functional	7,6,36,16	37,27,35,33,31,29,9,12,14	0	0	0	0	0	0	0	0	1	16
M17	Functional	7,37,6,36,35,33,31,29,9,12,14	27,16	1	0	0	0	0	0	0	0	1	16
M18	Functional	7,6,27	37,36,35,33,31,28,9,12,14,16	0	0	0	0	0	0	0	0	0	16
M19	Functional	7,6,35,33,31,29,9,12,14,16	37,36,27	0	0	0	0	0	0	0	0	0	16
M20	Functional	7,6,27,35,33,31,29,9,12,14,16	37,36	0	1	1	1	1	1	1	1	1	16
M21	Functional	7,6	37,36,27,35,33,31,29,9,12,14,16	0	1	1	1	1	1	1	1	1	16

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NOTES:

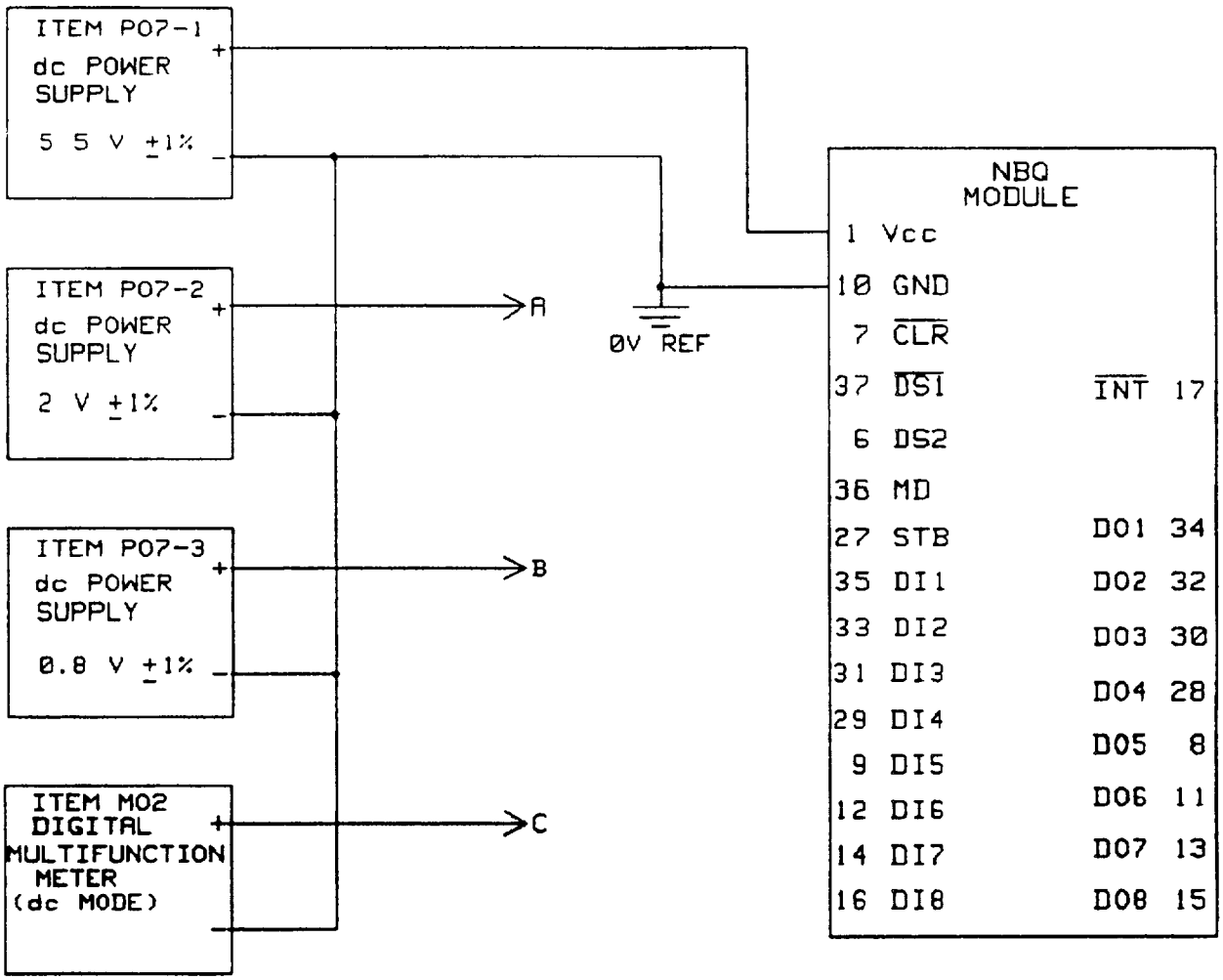
1. Only one pin under test at a time.

2. "0" = Vol = 0.45 V maximum.
   "1" = Voh = 3.5 V minimum.

3. Control input pins (6, 7, 27, 36, 37) shall be changed first for each test condition.

4. The functional pattern sequence is to be run twice, once at Vcc = 4.5 V and once at Vcc = 5.5 V level.

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NOTE:

1. Load all outputs as shown on figure 14.

FIGURE 16. Functional test circuit for table XIII.

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TABLE XIV. Propagation delay time (tPHL) test.

Test cond	Parameter	Connect A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	25 ± 5°C Initial limits Min	Max	Units	Figs
N1	DI1 to DO1	7,36	Open	35	6,27	37	34	5	32	nS	17,18
N2	DI2 to DO2			33			32				
N3	DI3 to DO3			31			30				
N4	DI4 to DO4			29			28				
N5	DI5 to DO5			9			8				
N6	DI6 to DO6			12			11				
N7	DI7 to DO7			14			13				
N8	DI8 to DO8	7,36	Open	16	6,27	37	15	5	32	nS	17,18
N9	STB to DO1	6,7	36,37	35	27	Open	34	5	42	nS	17,19
N10	STB to DO2			33			32				
N11	STB to DO3			31			30				
N12	STB to DO4			29			28				
N13	STB to DO5			9			8				
N14	STB to DO6			12			11				
N15	STB to DO7			14			13				
N16	STB to DO8	6,7	36,37	16	27	Open	15				
N17	$\overline{DSI}$ to DO1	6,7,36	Open	35	Open	37	34				
N18	$\overline{DSI}$ to DO2			33			32				
N19	$\overline{DSI}$ to DO3			31			30				
N20	DSI to DO4			29			28				
N21	$\overline{DSI}$ to DO5			9			8				
N22	$\overline{DSI}$ to DO6	6,7,36	Open	12	Open	37	11	5	42	nS	17,19

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TABLE XIV. Propagation delay time (tPHL) test - Continued.

Test cond	Parameter	Connect A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	25 ± 5°C Initial limits Min	Max	Units	Figs
N23	DST to D07	6,7,36	Open	14	Open	37	13	5	42	nS	17,19
N24	DST to D08	6,7,36	Open	18	Open	37	15				
N25	DS2 to D01	7,36	37	35	5	Open	34				
N26	DS2 to D02			33			32				
N27	DS2 to D03			31			30				
N28	DS2 to D04			29			28				
N29	DS2 to D05			9			8				
N30	DS2 to D06			12			11				
N31	DS2 to D07			14			13				
N32	DS2 to D08	7,36		16	6		15	5	42	nS	17,19
N33	CLR to D01	27,36, 35,33, 31,29, 14,16, 9,12		6	7		34	5	57	nS	17,20
N34	CLR to D02						32				
N35	CLR to D03						30				
N36	CLR to D04						28				
N37	CLR to D05						8				
N38	CLR to D06						11				
N39	CLR to D07						13				
N40	CLR to D08	27,36, 35,33, 31,29, 14,16, 9,12		6	7		15	5	57	nS	17,20
N41	STB to INT	7	37	27	6	Open	17	5	42	nS	17,21

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Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (tPLH) test.

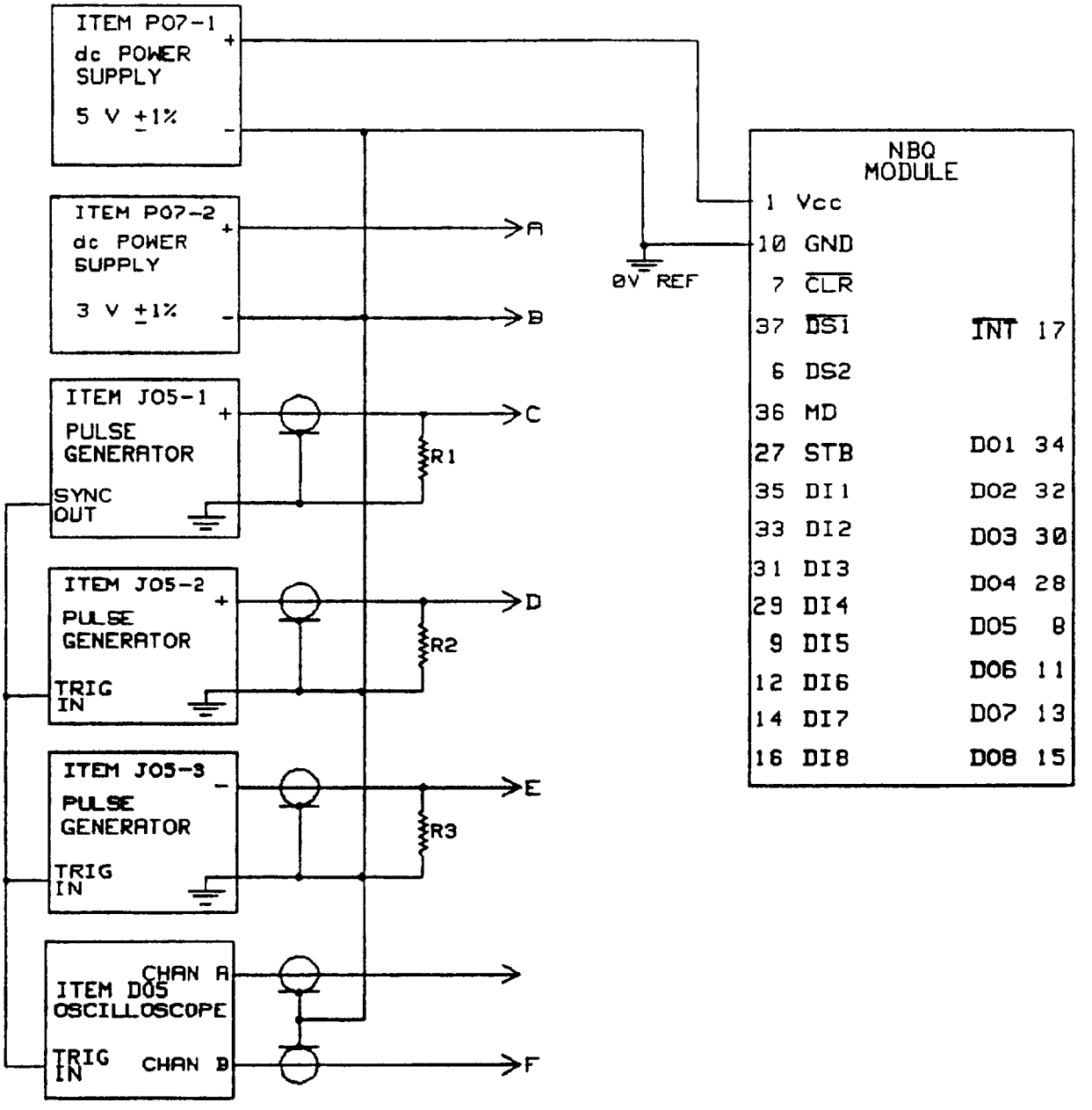
Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
P1	DI1 to DO1	7,36	Open	35	6,27	37	34	5	32	nS	17,22
P2	DI2 to DO2			33			32				
P3	DI3 to DO3			31			30				
P4	DI4 to DO4			29			28				
P5	DI5 to DO5			9			8				
P6	DI6 to DO6			12			11				
P7	DI7 to DO7			14			13				
P8	DI8 to DO8	7,36	Open	16	6,27	37	15	5	32	nS	17,22
P9	STB to DO1	6,7	36,37	35	27	Open	34	5	42	nS	17,19
P10	STB to DO2			33			32				
P11	STB to DO3			31			30				
P12	STB to DO4			29			28				
P13	STB to DO5			9			8				
P14	STB to DO6			12			11				
P15	STB to DO7			14			13				
P16	STB to DO8	6,7	36,37	16	27	Open	15				
P17	DST to DO1	6,7,36	Open	35	Open	37	34				
P18	DST to DO2			33			32				
P19	DST to DO3			31			30				
P20	DST to DO4			29			28				
P21	DST to DO5			9			8				
P22	DST to DO6	6,7,36	Open	12	Open	37	11	5	42	nS	17,19

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TABLE XV. Propagation delay time (tPLH) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	25 + 5°C Initial T limits Min	25 + 5°C Initial T limits Max	Units	Figs
P23	$\overline{DS1}$ to D07	6,7,36	Open	14	Open	37	13	5	42	nS	17,19
P24	$\overline{DS1}$ to D08	6,7,36	Open	16	Open	37	15				
P25	DS2 to D01	7,36	37	35	6	Open	34				
P26	DS2 to D02			33			32				
P27	DS2 to D03			31			30				
P28	DS2 to D04			29			28				
P29	DS2 to D05			9			8				
P30	DS2 to D06			12			11				
P31	DS2 to D07			14			13				
P32	DS2 to D08	7,36		18			15	5	42	nS	17,19
P33	$\overline{DS2}$ to $\overline{INT}$	7	37	27	6	Open	17	5	32	nS	17,21
P34	$\overline{DS1}$ to $\overline{INT}$	6,7	Open	27	Open	37	17	5	32	nS	17,21

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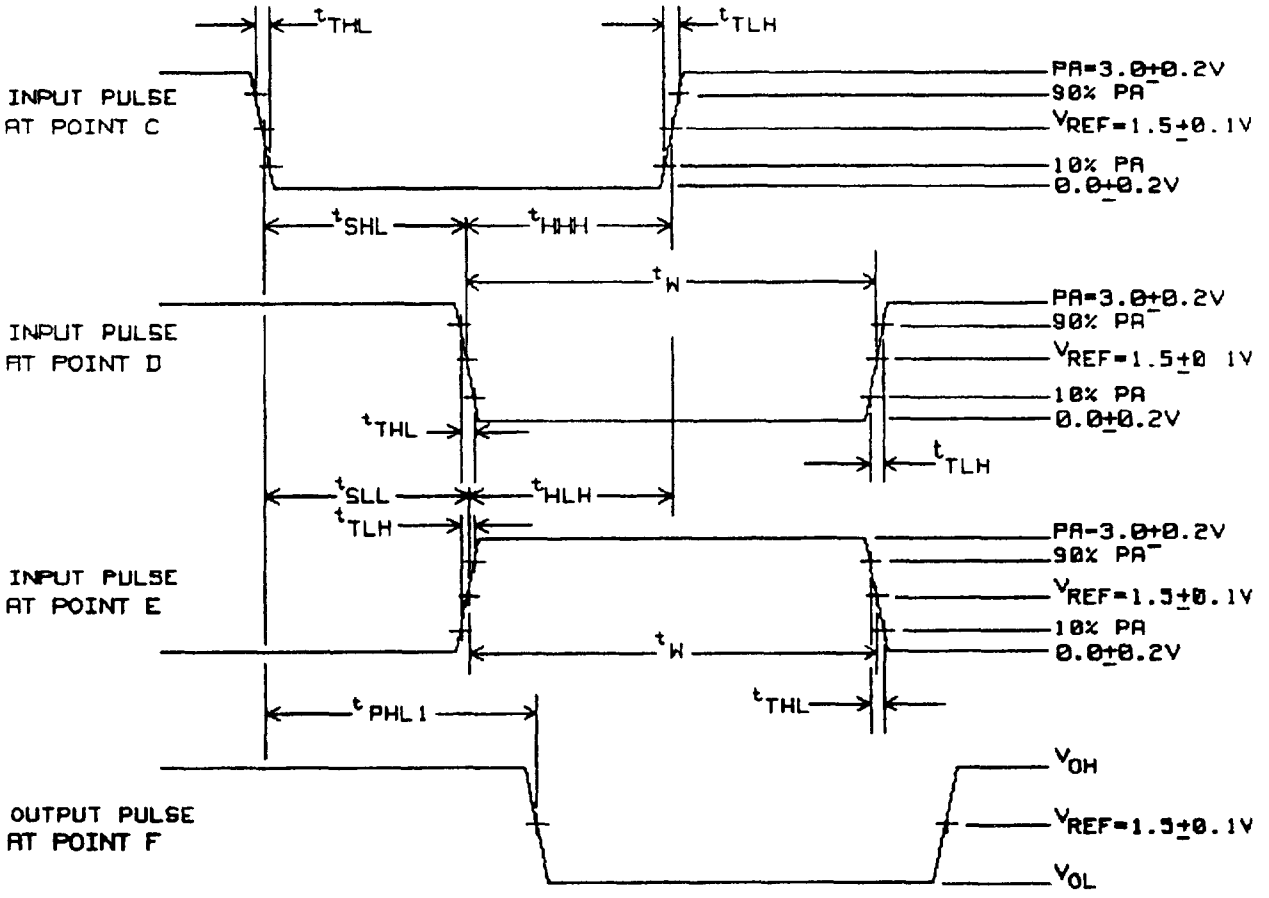


NOTES:

1. R1 = R2 = R3 = 51 Ohm ± 5 percent, 1/2 W.
2. Load all outputs as shown on figure 14.

FIGURE 17. Propagation delay time test circuit for tables XIV and XV.

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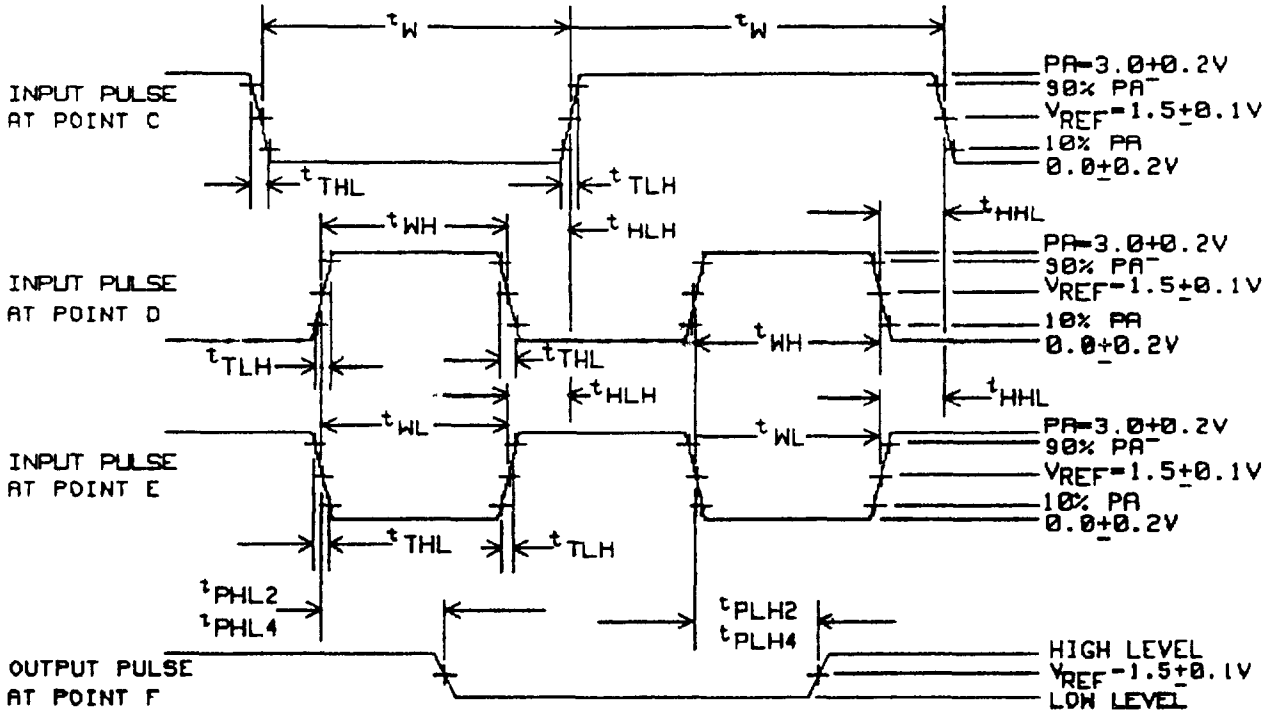
NOTE:

1. Input pulse parameters:

   Cycle time = 500 ± 50 nS.
   tW = 60 ± 5 nS.
   tSHL = tSLL = 20, +2, -0 nS.
   tHHL = tHLH = 30, +2, -0 nS.
   tTLH = tTHL = 8 ± 4 nS.

FIGURE 18. Propagation delay time (tPHL) test waveforms for table XIV.

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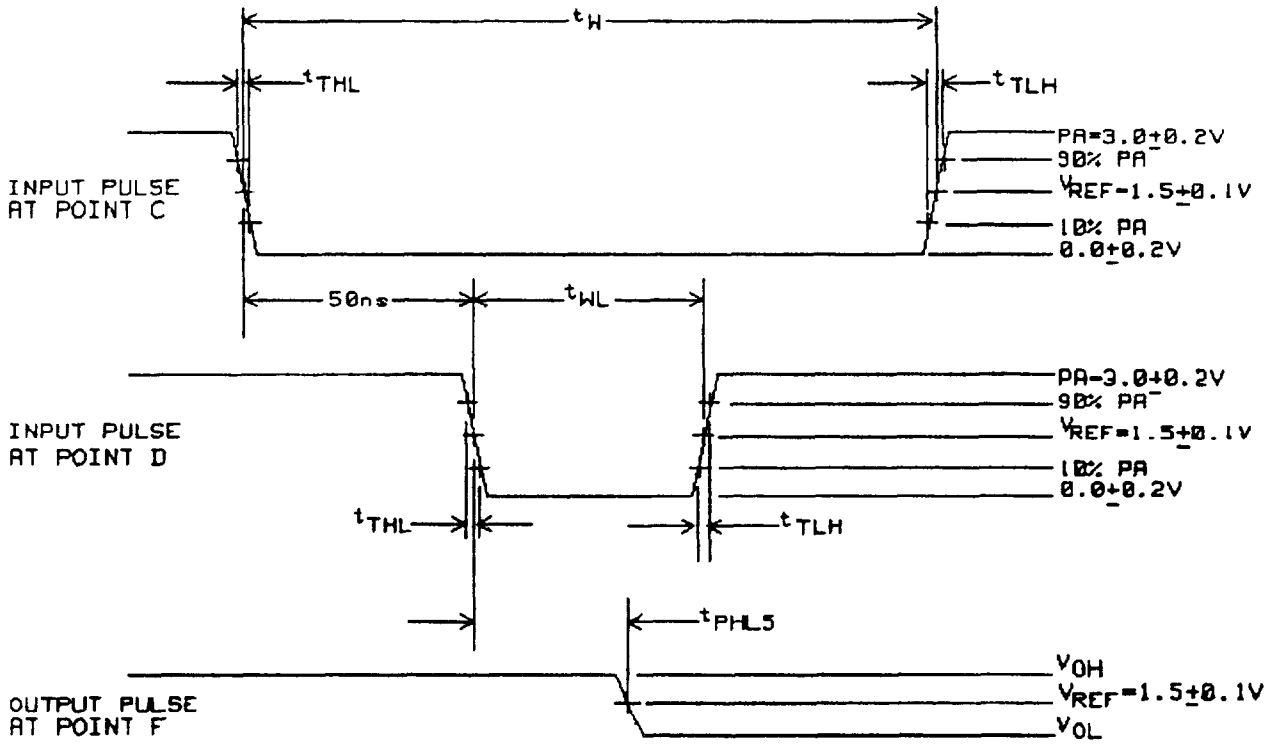
NOTE:

1. **Input pulse parameters:**

   tW = 125 + 10, -0 nS.
   tWH = tWL = 40 + 2, -0 nS.
   tHLH = tHHL = 30 + 2, -0 nS.
   tTLH = tTHL = 8 ± 4 nS.

FIGURE 19. Propagation delay time (tPHL and tPLH) test waveforms for tables XIV and XV.

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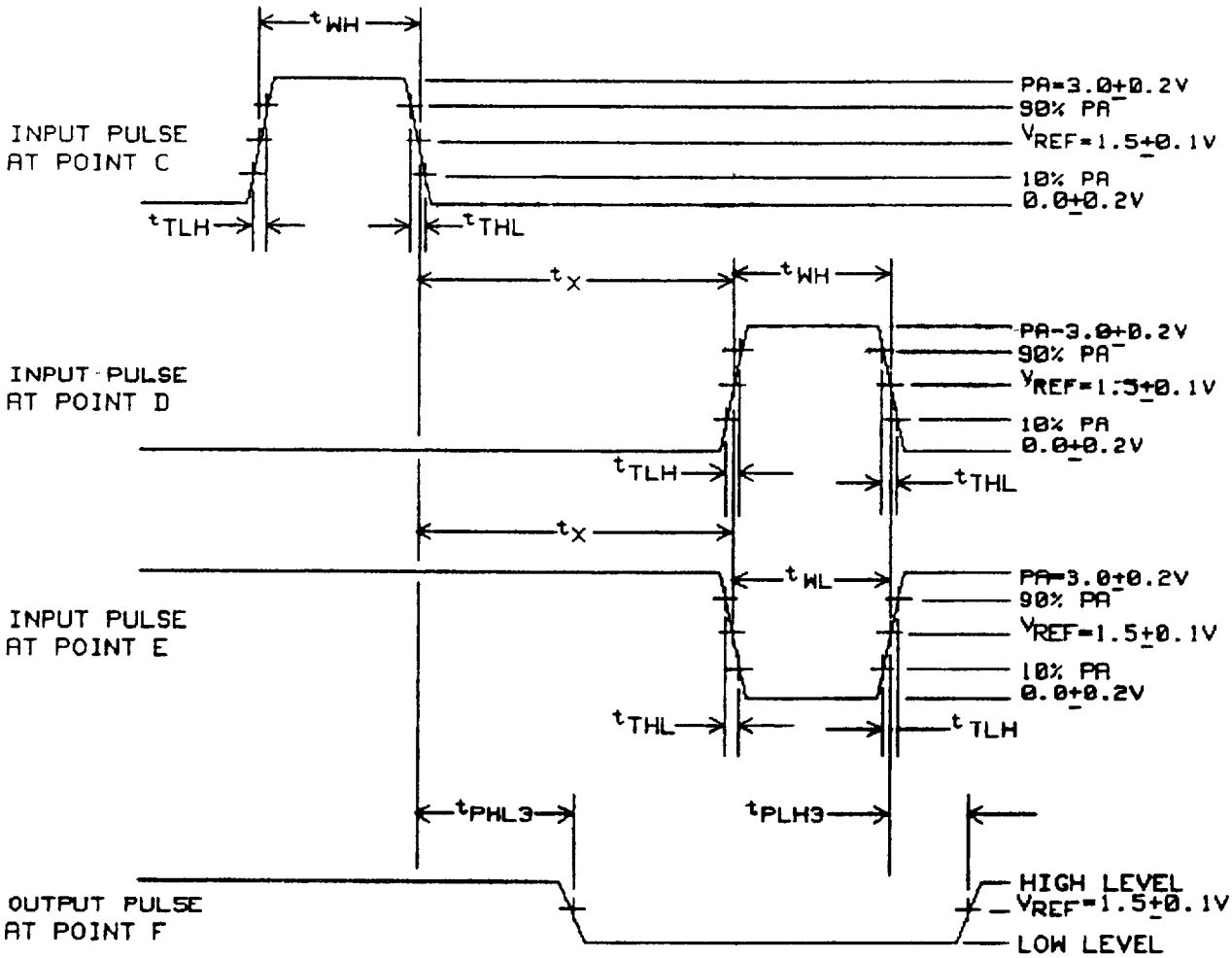
NOTE:

1. Input pulse parameters:

   Cycle time = 500 ± 50 nS.
   tW = 125 + 10, -0 nS.
   tWL = 40 + 2, -0 nS.
   tTLH = tTHL = 8 ± 4 nS.

FIGURE 20. Propagation delay time (tPHL) test waveforms for table XIV.

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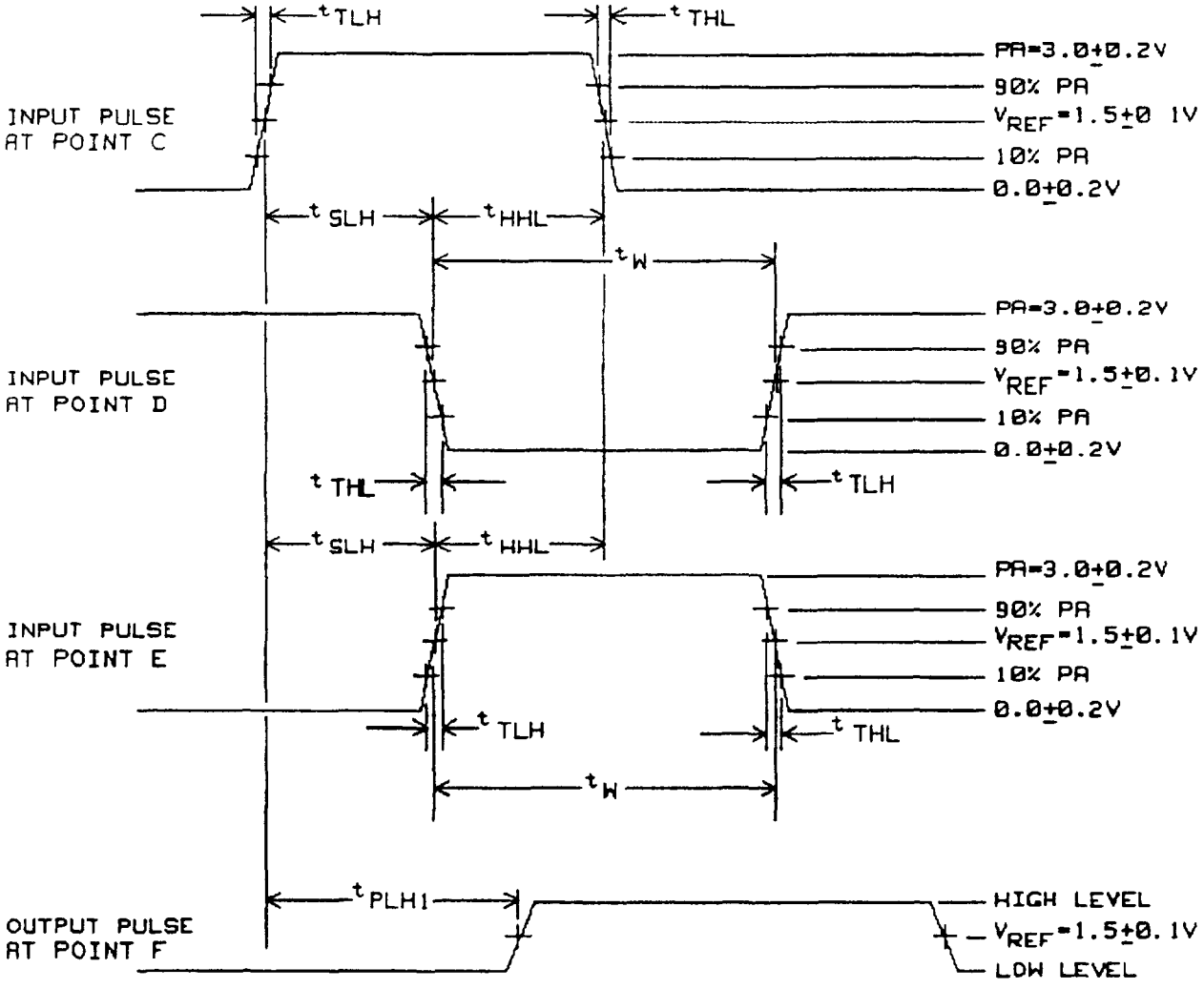
NOTE:

1. Input pulse parameters:

   Cycle time = 500 ± 50 nS.
   tWH = tWL = 40 + 2, -0 nS.
   tX = 105 ± 5 nS.
   tTLH = tTHL = 8 ± 4 nS.

FIGURE 21. Propagation delay time (tPHL and tPLH) test waveforms for tables XIV and XV.

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NOTE:

1. Input pulse parameters:

   Cycle time = 500 ± 50 nS.
   tW = 50 + 5, -0 nS.
   tSLH = 20 + 2, -0 nS.
   tHHL = 30 + 2, -0 nS.
   tTLH = tTHL = 8 ± 4 nS.

FIGURE 22. Propagation delay time (tPLH) test waveforms for table XV.

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TABLE XVI. Disable delay time (tPHZ) test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Figs
Q1	$\overline{DST}$ to DO1	6,7,27,35,33,31,29,9,12,14,16	36	37	34	5	150	nS	23,24
Q2	$\overline{DS1}$ to DO2				32				
Q3	$\overline{DST}$ to DO3				30				
Q4	$\overline{DST}$ to DO4				28				
Q5	$\overline{DST}$ to DO5				8				
Q6	$\overline{DS1}$ to DO6				11				
Q7	$\overline{DST}$ to DO7				13				
Q8	$\overline{DST}$ to DO8	6,7,27,35,33,31,29,9,12,14,16	36	37	15				
Q9	DS2 to DO1	27,7,35,33,31,29,9,12,14,16	36,37	6	34	5	150	nS	23,24
Q10	DS2 to DO2				32				
Q11	DS2 to DO3				30				
Q12	DS2 to DO4				28				
Q13	DS2 to DO5				8				
Q14	DS2 to DO6				11				
Q15	DS2 to DO7	27,7,35,33,31,29,9,12,14,16	36,37	6	13	5	150	nS	23,24

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TABLE XVI. Disable delay time (tPHZ) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
Q16	DS2 to D08	27,7,35,33,31, 29,9,12,14,16	36,37	6	15	5	150	nS	23,24
Q17	MD to D01		37,6	36	34				
Q18	MD to D02				32				
Q19	MD to D03				30				
Q20	MD to D04				28				
Q21	MD to D05				8				
Q22	MD to D06				11				
Q23	MD to D07				13				
Q24	MD to D08	27,7,35,33,31, 29,9,12,14,16	37,6	36	15	5	150	nS	23,24

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:51Z
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TABLE XVII. Disable delay time (tPLZ) test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
R1	$\overline{DS1}$ to DO1	6,27,7	36,35,33,31,29,9,12,14,16	37	34	5	60	nS	23,24
R2	$\overline{DS1}$ to DO2	"	"	"	32	"	"	"	"
R3	$\overline{DS1}$ to DO3	"	"	"	30	"	"	"	"
R4	$\overline{DS1}$ to DO4	"	"	"	28	"	"	"	"
R5	$\overline{DS1}$ to DO5	"	"	"	8	"	"	"	"
R6	$\overline{DS1}$ to DO6	"	"	"	11	"	"	"	"
R7	$\overline{DS1}$ to DO7	"	"	"	13	"	"	"	"
R8	$\overline{DS1}$ to DO8	8,27,7	36,35,33,31,29,9,12,14,16	37	15	"	"	"	"
R9	DS2 to DO1	27,7	36,37,35,33,31,29,9,12,14,16	6	34	"	"	"	"
R10	DS2 to DO2	"	"	"	32	"	"	"	"
R11	DS2 to DO3	"	"	"	30	"	"	"	"
R12	DS2 to DO4	"	"	"	28	"	"	"	"
R13	DS2 to DO5	"	"	"	8	"	"	"	"
R14	DS2 to DO6	"	"	"	11	"	"	"	"
R15	DS2 to DO7	27,7	36,37,35,33,31,29,9,12,14,16	6	13	5	60	nS	23,24

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TABLE XVII. Disable delay time (tPLZ) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
R16	DS2 to D08	27,7	36,37,35,33,31, 29,9,12,14,16	6	15	5	60	nS	23,24
R17	MD to D01		37,6,35,33,31, 29,9,12,14,16	36	34				
R18	MD to D02				32				
R19	MD to D03				30				
R20	MD to D04				28				
R21	MD to D05				8				
R22	MD to D06				11				
R23	MD to D07				13				
R24	MD to D08	27,7	37,6,35,33,31, 29,9,12,14,16	36	15	5	60	nS	23,24

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:51Z
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TABLE XVIII. Enable delay time (tPZL) test

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Figs
S1	$\overline{DS1}$ to DO1	6,27,7	36,35,33,31,29,9,12,14,16	37	34	5	60	nS	23,24
S2	$\overline{DS1}$ to DO2				32				
S3	$\overline{DS1}$ to DO3				30				
S4	$\overline{DS1}$ to DO4				28				
S5	$\overline{DS1}$ to DO5				8				
S6	$\overline{DS1}$ to DO6				11				
S7	$\overline{DS1}$ to DO7				13				
S8	$\overline{DS1}$ to DO8	6,27,7	36,35,33,31,29,9,12,14,16	37	15				
S9	DS2 to DO1	27,7	36,37,35,33,31,29,9,12,14,16	6	34				
S10	DS2 to DO2				32				
S11	DS2 to DO3				30				
S12	DS2 to DO4				28				
S13	DS2 to DO5				8				
S14	DS2 to DO6				11				
S15	DS2 to DO7	27,7	36,37,35,33,31,29,9,12,14,16	6	13	5	60	nS	23,24

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TABLE XVIII. Enable delay time (tPZL) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Figs
S16	DS2 to D08	27,7	36,37,35,33,31, 29,9,12,14,16	6	15	5	60	nS	23,24
S17	MD to D01	6,27, 7	37,6,35,33,31, 29,9,12,14,16	36	34				
S18	MD to D02				32				
S19	MD to D03				30				
S20	MD to D04				28				
S21	MD to D05				8				
S22	MD to D06				11				
S23	MD to D07				13				
S24	MD to D08	6,27, 7	37,6,35,33,31, 29,9,12,14,16	36	15	5	60	nS	23,24

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TABLE XIX. Enable delay time (tPZH) test.

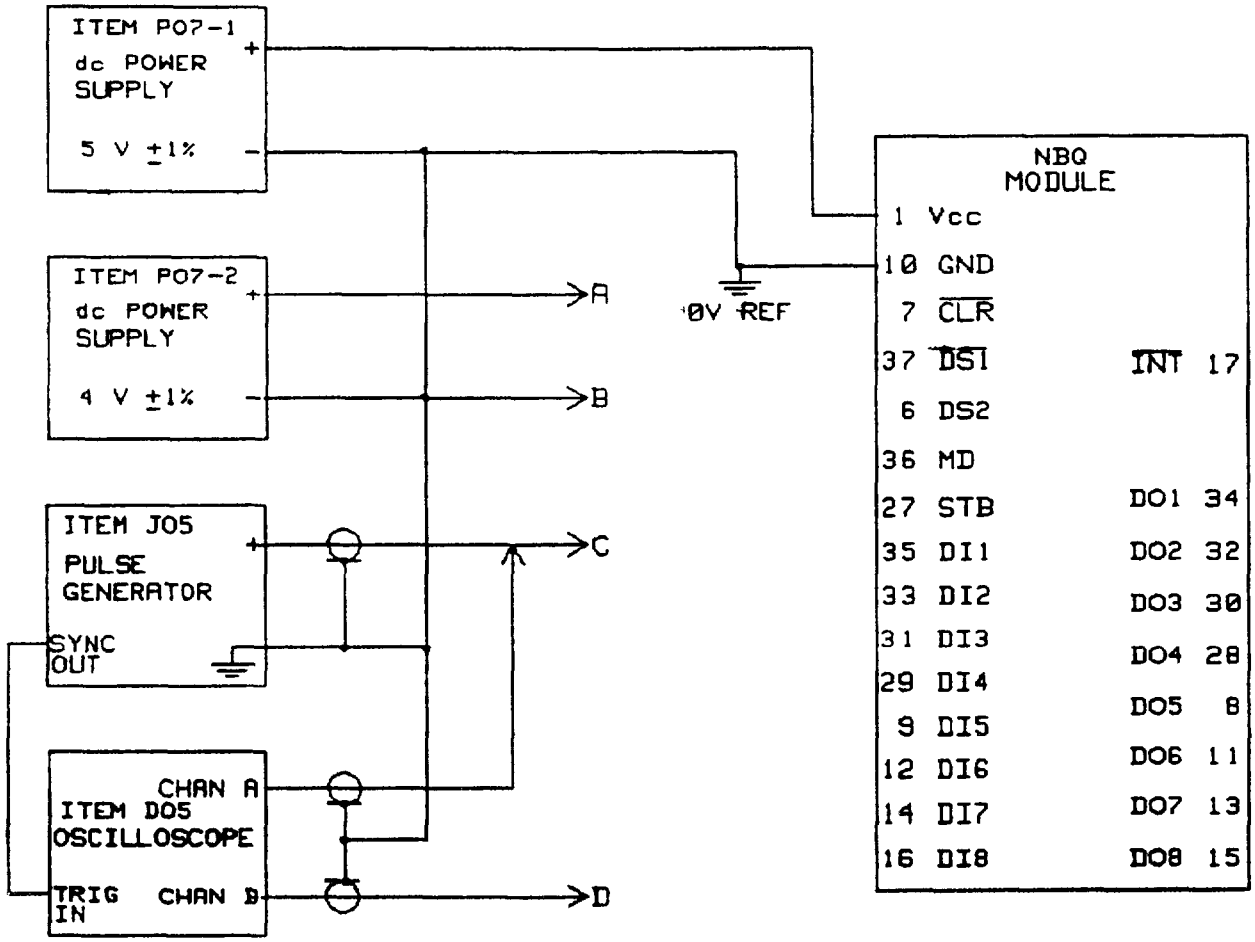
Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Figs
T1	$\overline{DST}$ to DO1	6,7,27,35,33, 31,29,9,12,14, 16	36	37	34	5	60	nS	23,24
T2	$\overline{DST}$ to DO2				32				
T3	$\overline{DST}$ to DO3				30				
T4	$\overline{DST}$ to DO4				28				
T5	$\overline{DST}$ to DO5				8				
T6	$\overline{DST}$ to DO6				11				
T7	$\overline{DST}$ to DO7				13				
T8	$\overline{DST}$ to DO8	6,7,27,35,33, 31,29,9,12,14, 16	36	37	15				
T9	DS2 to DO1	27,7,35,33,31, 29,9,12,14,16	36,37	6	34				
T10	DS2 to DO2				32				
T11	DS2 to DO3				30				
T12	DS2 to DO4				28				
T13	DS2 to DO5				8				
T14	DS2 to DO6				11				
T15	DS2 to DO7	27,7,35,33,31, 29,9,12,14,16	36,37	6	13	5	60	nS	23,24

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TABLE XIX. Enable delay time (tPZH) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Figs
T16	DS2 to DO8	27,7,35,33,31, 29,9,12,14,16	36,37	6	15	5	60	nS	23,24
T17	MD to DO1		37,6	36	34				
T18	MD to DO2				32				
T19	MD to DO3				30				
T20	MD to DO4				28				
T21	MD to DO5				8				
T22	MD to DO6				11				
T23	MD to DO7				13				
T24	MD to DO8	27,7,35,33,31, 29,9,12,14,16	37,6	36	15	5	60	nS	23,24

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:51Z
Check the source to verify that this is the current version before use.

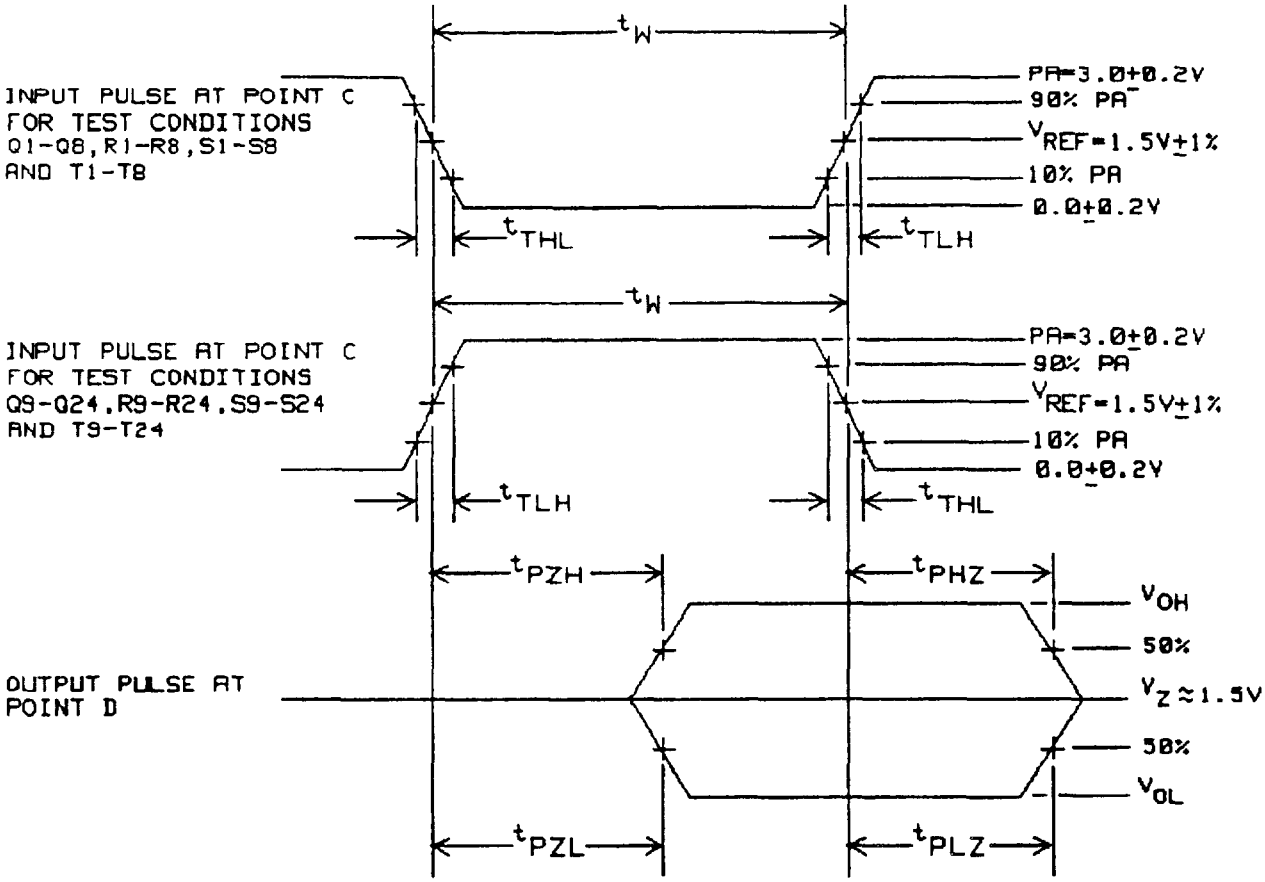


NOTE:

1. Load outputs as shown on figure 14, for test conditons Q1-Q24, R1-R24, S1-S24 and T1-T24 and W1-W9.

FIGURE 23. Disable/Enable delay time, output crosstalk (low and high level), output crosstalk duration, and power supply transient amplitude test circuit for tables XVI, XVII, XVIII, XIX, XX, XXI, XXII, and XXIII.

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NOTES:

1. Input pulse parameters:

   Cycle time = 1000 ± 50 nS.
   tW = 250 ± 25 nS.
   tTLH = tTHL = 8 ± 4 nS.

2. Measure at 50 percent point between logic level and high impedance level on outputs.

FIGURE 24. Disable/Enable delay time test waveforms for tables XVI, XVII, XVIII and XIX.

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TABLE XX. Output crosstalk, low level, test.

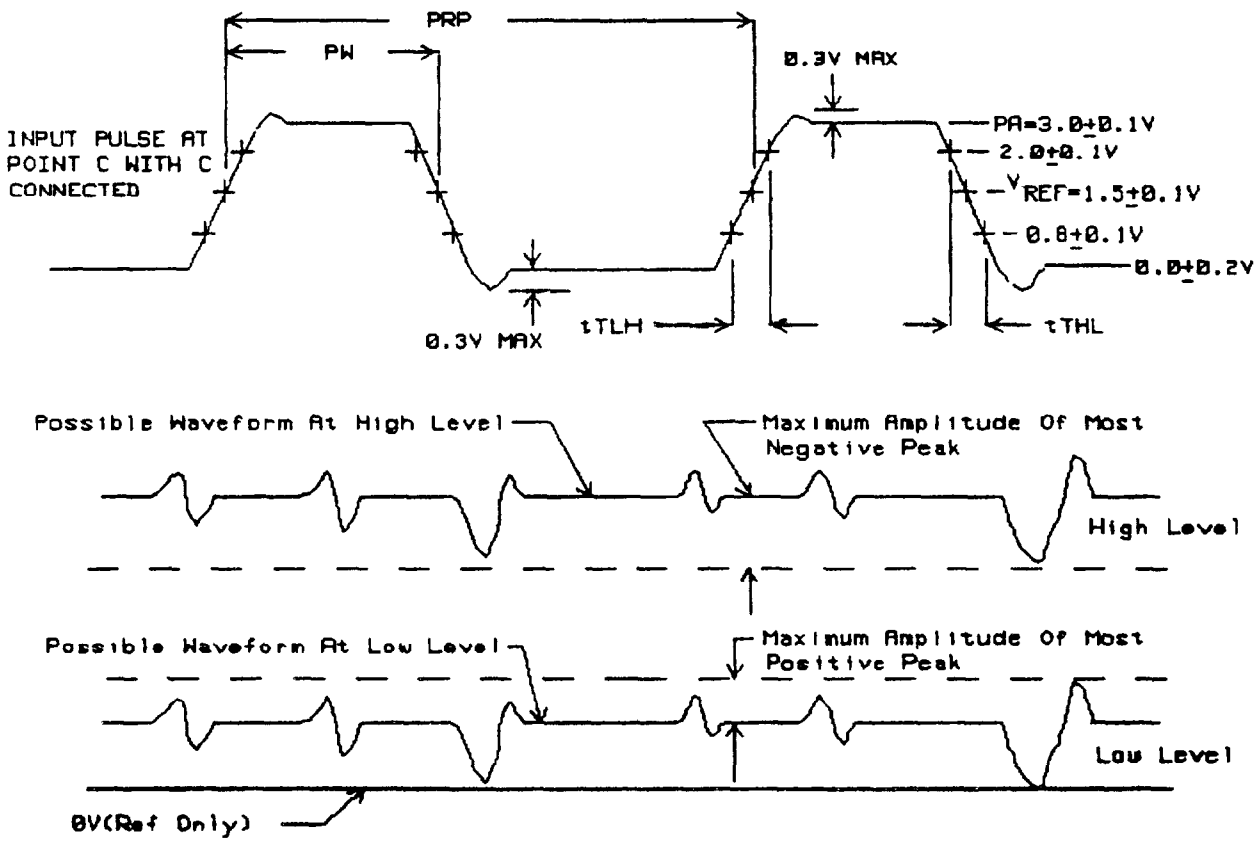
Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
U1	$\overline{INT}$	7,6, 27,36	37	35,33,31, 29,9,12, 14,16	17	0	950	mV	23,25
U2	DO1		35	33,31,29, 9,12,14, 16,37	34				
U3	DO2		33	35,31,29, 9,12,14, 16,37	32				
U4	DO3		31	35,33,29, 9,12,14, 16,37	30				
U5	DO4		29	35,33,31, 9,12,14, 16,37	28				
U6	DO5		9	35,33,31, 29,12,14, 16,37	8				
U7	DO6		12	35,33,31, 29,9,14, 16,37	11				
U8	DO7		14	35,33,31, 29,9,12, 16,37	13				
U9	$\overline{DO8}$	7,6, 27,36	16	35,33,31, 29,9,12, 14,37	15	0	950	mV	23,25

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TABLE XXI. Output crosstalk, high level, test

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Figs
V1	$\overline{INT}$	7,6,27, 36,37	Open	35,33,31, 29,9,12, 14,16	17	0	400	mV	23,25
V2	DO1	7,6,27, 36,35		33,31,29, 9,12,14, 16,37	34				
V3	DO2	7,6,27, 36,33		35,31,29, 9,12,14, 16,37	32				
V4	DO3	7,6,27, 36,31		35,33,29, 9,12,14, 16,37	30				
V5	DO4	7,6,27, 36,29		35,33,31, 9,12,14, 16,37	28				
V6	DO5	7,6,27, 36,9		35,33,31, 29,12,14, 16,37	8				
V7	DO6	7,6,27, 36,12		35,33,31, 29,9,14, 16,37	11				
V8	DO7	7,6,27, 36,14		35,33,31, 29,9,12, 16,37	13				
V9	DO8	7,6,27, 36,16	Open	35,33,31, 29,9,12, 14,37	15	0	400	mV	23,25

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:51Z
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NOTE:

1. Input pulse parameters:

   PW = 500 ± 50 nS.
   PRP = 1000 ± 100 nS.
   tTLH = tTHL = 15 ± 3 nS.

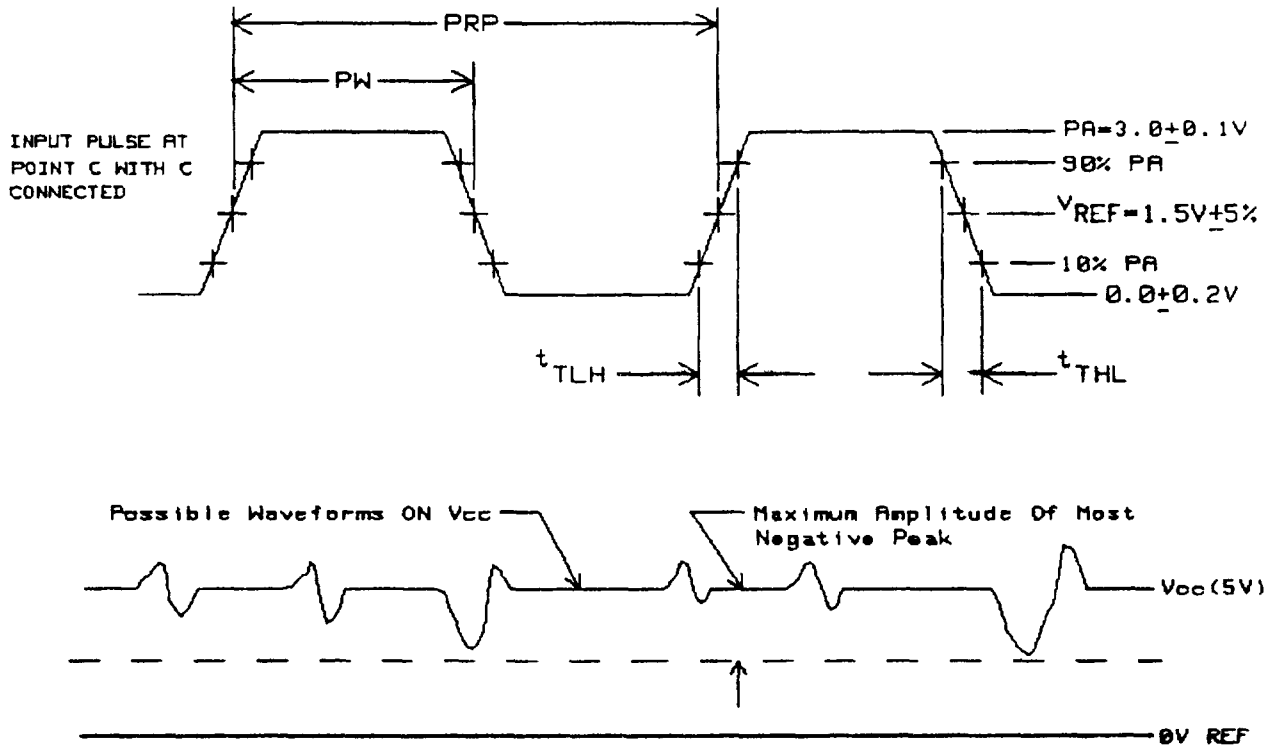
FIGURE 25. Output crosstalk, low and high level, waveforms for tables XX and XXI.

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TABLE XXII. Power supply transient amplitude test.

Test cond	Parameter	Connect				25 ± 5°C Initial limits		Units	Figs
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W	PSTA	7,6, 27, 36	37	35,33, 31,29, 14,16, 12,9	1	0	400	mV	23, 26

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NOTE:

1. Input pulse parameters:

   PW = 500 ± 50 nS.
   PRP = 1000 ± 100 nS.
   tTLH = tTHL = 10 ± 2 nS.

FIGURE 26. Power supply transient amplitude waveform for table XXII.

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Check the source to verify that this is the current version before use.

TABLE XXIII. Power supply filter capacitance test.

Test cond	Parameter	Connect		25 ± 5°C Initial limits		Units	Fig
		A to pin	B to pin	Min	Max		
X	PSFC	Open	1	3.3	6.1	uF	27

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:51Z
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TABLE XXIV. Input capacitance test.

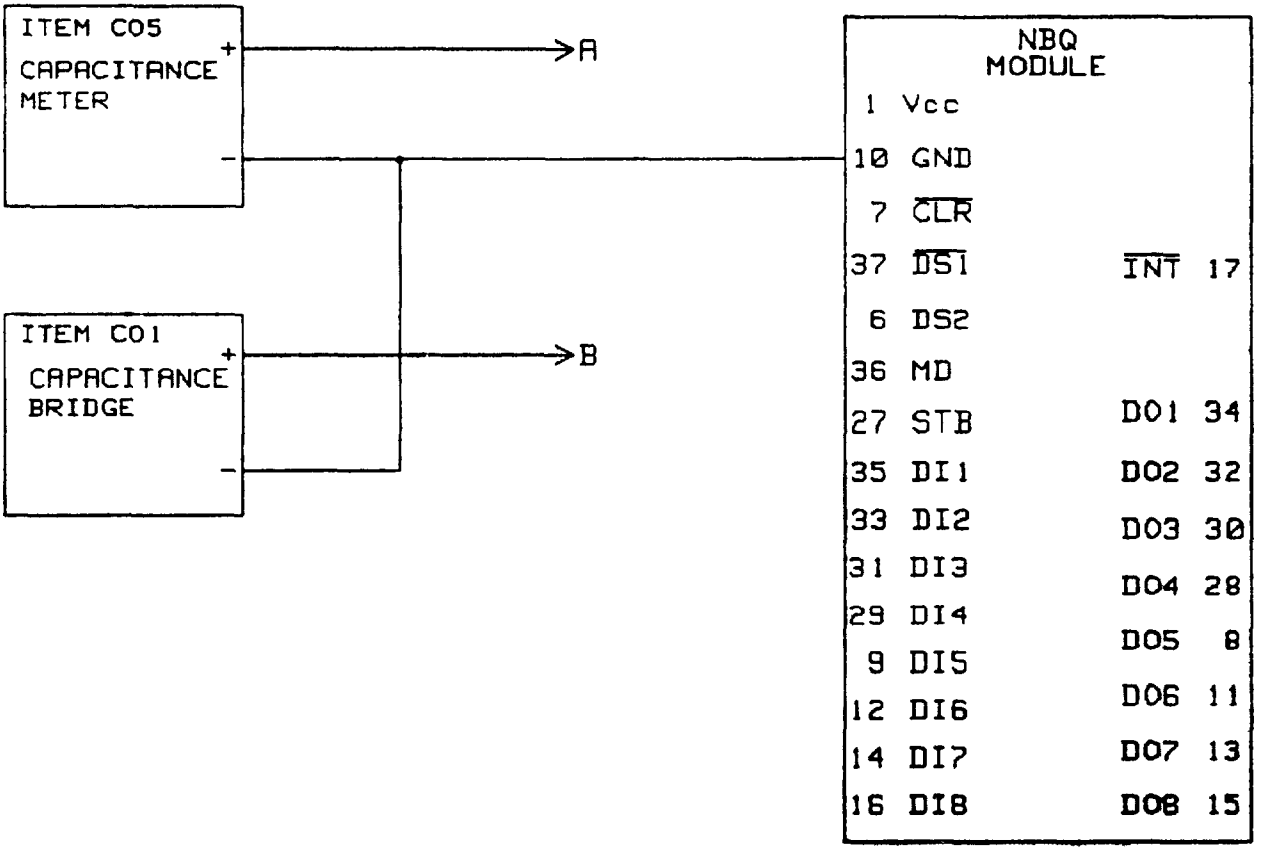
Test cond	Signal name	Connect A to no.	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
Y1	MD	36	Open	2	18	pF	27
Y2	$\overline{DS1}$	37		2	18	pF	27
Y3	DS2	6		2	15	pF	27
Y4	$\overline{CLR}$	7					
Y5	STB	27					
Y6	DI1	35					
Y7	DI2	33					
Y8	DI3	31					
Y9	DI4	29					
Y10	DI5	9					
Y11	DI6	12					
Y12	DI7	14					
Y13	DI8	16	Open	2	15	pF	27

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TABLE XXV. Output capacitance test.

Test cond	Signal name	Connect A to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
Z1	DO1	34	2	18	pF	27
Z2	DO2	32				
Z3	DO3	30				
Z4	DO4	28				
Z5	DO5	8				
Z6	DO6	11				
Z7	DO7	13				
Z8	DO8	15	2	18	pF	27

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NOTES:

1. Null out the effects of the test leads prior to connecting the module.

2. Do not attach test connector or wires to any of the module pins other than pin 10 and pin under test. Stray capacitance added to other pins may affect the reading.

3. Capacitance bridge (C01) setup: Frequency = 120 Hz, Vpk = 0.3 V maximum.

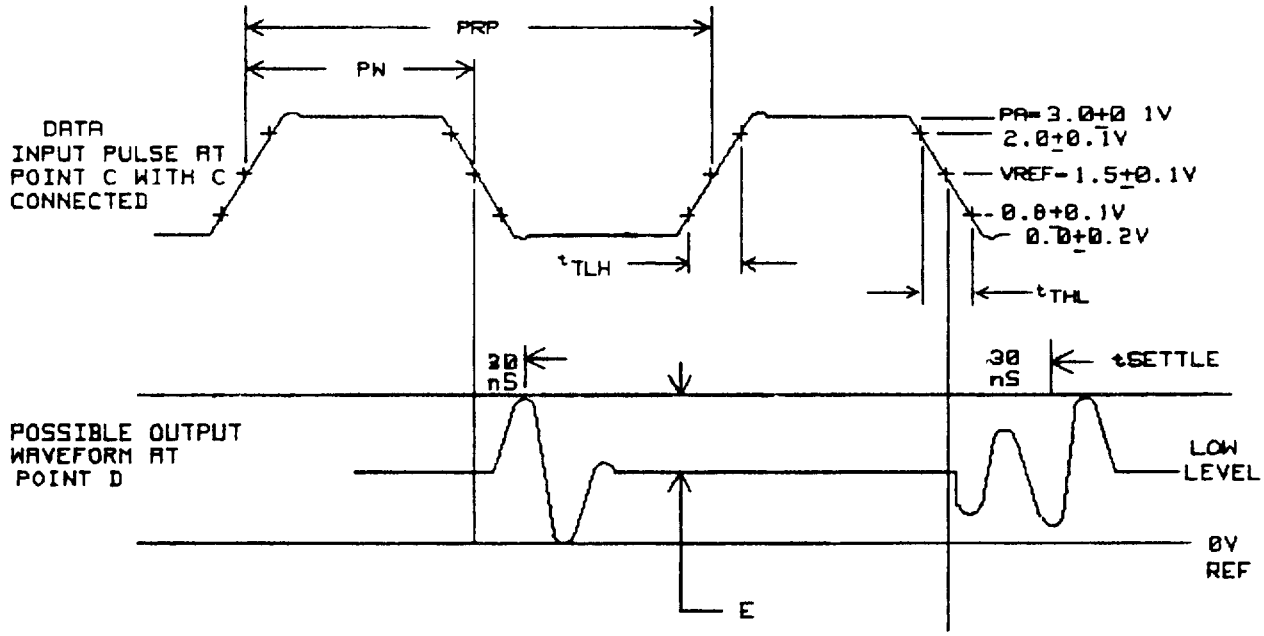
FIGURE 27. Power supply filter, input and output capacitance test circuit for tables XXIII, XXIV and XXV.

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TABLE XXVI. Output crosstalk duration, low level, test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 + 5°C Initial T limits Min	25 + 5°C Initial T limits Max	Units	Figs
AA1	$\overline{INT}$	7,6, 27,36	37	35,33,31, 29,9,12, 14,16	17	0	400	mV	23,28
AA2	DO1		35	33,31,29, 9,12,14, 16,37	34				
AA3	DO2		33	35,31,29, 9,12,14, 16,37	32				
AA4	DO3		31	35,33,29, 9,12,14, 16,37	30				
AA5	DO4		29	35,33,31, 9,12,14, 16,37	28				
AA6	DO5		9	35,33,31, 29,12,14, 16,37	8				
AA7	DO6		12	35,33,31, 29,9,14, 16,37	11				
AA8	DO7		14	35,33,31, 29,9,12, 16,37	13				
AA9	DO8	7,6, 27,36	16	35,33,31, 29,9,12, 14,37	15	0	400	mV	23,28

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NOTES:

1. Data input parameters:

   PW = 500 ± 50 nS.
   PRP = 1000 ± 100 nS.
   tTLH = tTHL = 15 ± 3 nS.

2. E is the amplitude of the most positive peak of the waveform at point D measured after a settle time of 30 nS has elapsed. The settle time, tSETTLE, is measured beginning with the negative transition of the data input pulse.

FIGURE 28. Output crosstalk duration, low level, test waveform for table XXVI.

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